

INFLUENCE DYNAMICS IN HETEROGENEOUS LLM-BASED MULTI-AGENT SYSTEMS

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Abstract

Large Language Models (LLMs) are increasingly deployed in Multi-Agent Systems (MAS) where agents collaborate towards a shared solution. While prior work has shown that opinion dynamics in these systems are complex and can be misaligned with expectations, little focus has been paid to heterogeneous agent populations. In such systems, where the models differ in size, architecture and training data, it remains unclear how agents can influence each other’s outcomes, and whether a shift to a heterogeneous setting affects influence dynamics. We propose a framework for measuring influence dynamics in LLM-based MAS. An agent is tasked with a binary classification task in isolation, and is later asked to revisit this task, but now considering another agent’s opinion. Influence is measured as the shift in the agent’s certainty in its opinion before and after interaction. We evaluate several open-source models on three natural language understanding tasks. Results show that interaction alone produces frequent and substantial opinion shifts, even for models that are initially certain in their opinion. Furthermore, influence is driven mainly by the stability of the model being influenced and less by the influential power of the peer. Similarly, the transition from homogeneous to heterogeneous systems does not affect all models uniformly. Depending on the model, heterogeneity may either increase or decrease susceptibility. Overall, system composition, model types, and model roles are important factors in influence dynamics, and thus need to be considered when building robust MAS.¹

1 Introduction

Recent advancements of Large Language Models (LLMs) and improvement in their general-purpose capabilities have led them to their increasing integration into Multi-Agent Systems (MAS), where

these models are deployed to interact and collaborate with each other. The performance of such systems have enabled a shift from applying individual LLMs for problem-solving to the deployment of complex MAS in a wide range of domains and tasks, such as software engineering, healthcare, and simulation of human behaviours, amongst other (He et al., 2025; Salehi et al., 2025; Tran et al., 2025).

As these systems become more integrated into society, it becomes increasingly important to understand the dynamics in agent interaction. Prior work has shown that LLM agents may change their answers after observing peer responses, converge toward agreement in multi-agent debate, or become susceptible to peer pressure (Cho et al., 2025; Cau et al., 2025; Song et al., 2025). Other studies show that interaction alone introduce unexpected behaviour that is not visible when examining LLMs in isolation, including conversational bias, group-level bias, and performance degradation (Coppo-lillo et al., 2025; Ashery et al., 2025; Wynn et al., 2025).

The interaction between agents in MAS is shaped by several design choices, such as communication protocol, communication structure and agent capabilities (Guo et al., 2024). One important design choice concerns whether agents are instantiated from the same underlying model, resulting in a homogeneous system, or from different model types, resulting in a heterogeneous system. While most existing research study multi-agent interaction in homogeneous settings, real-world deployments are not necessarily limited to such configurations. In practice, MAS may combine agents that differ in model family, size, training data, and training methods creating heterogeneous populations of interacting agents (MoltBook, 2026; OpenClaw, 2026).

This misalignment between real-world deployments and research reveals a gap in our understand-

¹Code and data are publicly available at: <https://github.com/fridanl/InfluenceDynamicsMAS>.

ing of LLM-based MAS. If agents differ in their underlying model properties, they may also differ in how they interact with one another. Some models may be more effective at steering the system outcome towards their opinion, while others may be more vulnerable to peer opinions. For example, a model with a large parameter size may be more effective at influencing and more resistant to influence than a smaller model. Similarly, a model that is highly certain in an opinion may be less susceptible to peer influence than one with low certainty. Such heterogeneity is a critical consideration for system design and security, since a model’s behaviour in a homogeneous setup may not generalise to a system composed of distinct model types. If certain model types are systematically more susceptible to influence from others, this imbalance could be exploited e.g., by introducing a highly influential model to steer the outcome toward a particular opinion.

Characterising influence dynamics in heterogeneous MAS is therefore a necessary step toward building more robust and predictable systems. We introduce a framework that enables systematic measurement of the direction and magnitude of such influence. To obtain a direct and interpretable measure of influence, we define our framework around a bi-agent system, where agents interact on a binary classification task. A bi-agent setting allows us to isolate the behaviour of each model against a single peer. By evaluating all pairwise combinations across a set of models, we obtain a complete picture of the influence dynamics within the group. The binary classification setting makes influence directly measurable as a shift in the agent’s certainty between two possible labels. This allows us to attribute any shift in an agent’s opinion as influence toward or away from the peer’s opinion, without the added complexity introduced by multiple competing alternatives.

The framework we propose consists of two rounds. In the first round, each agent independently solves a binary classification task C times, establishing a baseline distribution over labels. In the second round, the agent is asked to revisit the same task conditioned on both its own previous answer and the answer of a peer agent. Influence is then measured as the shift in an agent’s label distribution before and after interaction with a peer. A shift towards the peer’s label is measured as positive influence and a shift away is measured as negative influence. We summarise these shifts over all in-

stances using two averaged normalised influence scores. The positive influence score ranges from 0 to 1 and measures the proportion of the possible range that a model has shifted toward the proposed label. A score of 1 indicates full adoption of the peer’s label, while a score of 0 indicates no influence. The negative influence score ranges from 0 to -1 and captures the proportion of the possible range that the model has shifted away from the proposed label. Here, 0 again indicates no influence, while -1 indicates full resistance to the peer’s proposal, with the model shifting entirely toward the opposite label of the peer.

Together, these two influence measures allow us to distinguish between adoption, stability and resistance in model interaction. From this, we examine whether influence varies systematically across heterogeneous model pairs and compare these dynamics to a homogeneous setup.

Using this framework, we study influence dynamics across seven open-source LLMs on three binary Natural Language Understanding (NLU) tasks. Our experiments show that influence is strong in disagreement across model pairs, with most positive influence scores reaching above 0.5, and negative influence scores reaching below -0.3 . Thus, on average when two models initially disagree, the influencing model is able to move the other model halfway toward a complete change of opinion. Adoption is not the only reaction to peer disagreement, the negative influence score show that some models react with resistance thus strengthening their initial opinion.

Furthermore, the dynamics are also directional. That is, a model that is highly effective at shifting others toward its opinion may itself be easily influenced when being exposed to peer input. In fact, the influence dynamics observed in the results are driven more by the stability of the model being influenced, and not the ability of the peer to exert influence.

Contrary to intuition, robustness to peer influence is not reflected in the models with very certain opinions. Our results show, that the models that predominantly hold consistent opinions are among the most susceptible to influence. This has direct implications for system design, as model selection based on individual performance alone may not predict how a model will behave when exposed to peer opinions.

The effect of transitioning from a homogeneous to a heterogeneous system on influence dynam-

ics is not uniform. In fact, it varies substantially across models and datasets, and either decreases or increases peer influence. For some models, heterogeneous interaction reduces susceptibility to influence, with one model showing a decrease of 79% relative to the influence in the homogeneous setting. This suggests, that the model is considerably less susceptible when interacting with a different model type than its own. For others the opposite holds, with an increase of positive influence up to 27%. This suggests that MAS stability is jointly determined by system composition and the specific characteristics of the receiving model.

2 Framework

We propose a framework for studying influence dynamics in LLM-based MAS, where agents are tasked to solve a binary classification task. The framework is structured as a simple multi-agent debate in which agents first make independent judgments and later reconsider their decisions after observing the opinion of another judge (Du et al., 2024). This setup is designed to isolate the effect of external information on an agent’s decision making.

Despite the simplicity of the system, measuring influence systematically is complicated due to two fundamental properties of LLM-based interaction. First, LLM text generation is stochastic, meaning an agent does not produce a single deterministic label for a given instance, but instead may produce varying labels, across repeated runs. Second, models may exhibit different label preferences prior to peer interaction, and an influence measure should therefore reflect the change in opinion with respect to the opinion it already holds. Accounting for both properties simultaneously requires a precise experimental design and a structured influence measure, which we explain in detail in the following sections.

In the **first round**, each agent is prompted to solve a binary task and return a label-explanation pair for the instance. This is repeated C times per instance, yielding a Bernoulli distribution over labels that capture the agent’s *baseline certainty*, reflecting its behaviour in the absence of external influence. We introduce repetitions of the same instance to account for the stochastic nature of LLM text generation.

In the **second round**, agents are placed in a pairwise interaction. We simulate a multi-agent debate

by prompting an agent to revisit its decision on the same instance, now conditioned on its own baseline opinion and the baseline opinion of a peer agent. Here, *baseline opinion* refers to a label-explanation pair produced by an agent in the first round.

We discriminate between the agent that is asked to revisit its decision, which is referred to as the *receiver*, and the agent contributing the external information, which is referred to as the *peer*. We measure a peer’s influence on a receiver, by comparing the receiver’s baseline label distribution to the label distribution after interaction. Thus, we measure how exposure to a peer opinion shifts a receiver’s baseline certainty. Additionally, this shift covers the specific direction of influence. That is, whether the receiver moves toward the label proposed by the peer or away from it. Any observed shift is attributed to the receiver being influenced by the peer and forms the basis of our influence score. The full framework is illustrated in Figure 1, and described in the following three subsections.

2.1 Baseline Behaviour

We consider K distinct models, where an agent a^k is instantiated from. Each agent is prompted to solve a binary task instance s_i independently C times, and return a label-explanation pair for each independent run:

$$o_{i,c}^k = (\hat{y}_{i,c}^k, e_{i,c}^k). \quad (1)$$

Here $o_{i,c}^k$ represents the opinion of a^k , where $\hat{y}_{i,c}^k$ is the predicted label and $e_{i,c}^k$ is the corresponding explanation. Figure 1a, illustrates the baseline behaviour procedure for a single receiver-peer pair a^k and a^j and instance s_i , where both agents produce outputs in isolation. For a single instance, the sequence of label-explanation pairs forms the agent’s baseline output O_i^k :

$$O_i^k = \{o_{i,c}^k; c = 1, \dots, C\}. \quad (2)$$

An agent’s *baseline certainty* of a given label $y \in \{0, 1\}$ for the instance s_i is computed from its baseline output O_i^k and is defined as:

$$p_i^k(y) = \frac{1}{C} \sum_{c=1}^C \mathbb{I}(\hat{y}_{i,c}^k = y), \quad (3)$$

where $\hat{y}_{i,c}^k$ denotes the label predicted in repetition c and $\mathbb{I}(\cdot)$ is the indicator function. Thus, each model has two baseline certainties for every instance, one for each label. When we refer to a model as having

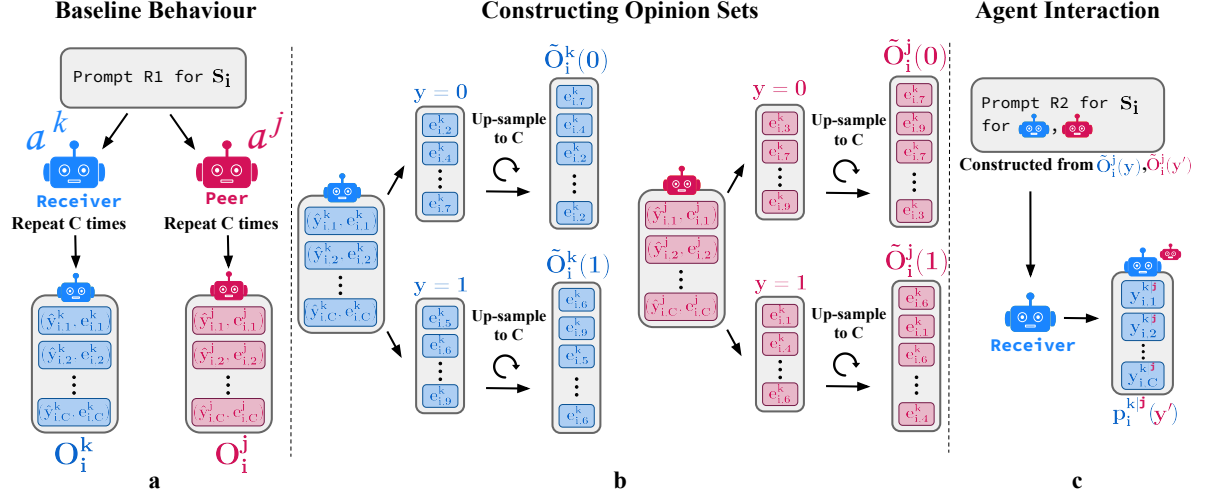


Figure 1: Overview of the framework. In the Baseline Behaviour panel (a), each agent solves the same task instance independently across C repetitions, producing a set of label-explanation pairs. In the Constructing Opinion Sets panel (b), these outputs are then separated by predicted label and up-sampled to construct label-stratified opinion sets of size C . Here both agents have produced inconsistent labels across the C repetitions in the Baseline Behaviour, so both opinion sets can be constructed for both agents. In Agent Interaction (c), the receiver is prompted to solve the task again while conditioned on its own fixed-label opinion and the peer’s fixed-label opinion.

a high baseline certainty, we mean it has fully consistent baseline predictions across the C repetitions, such that either $p_i^k(y=0) = 1$ or $p_i^k(y=1) = 1$.

2.2 Constructing Opinion Sets

Since an interaction between two agents is simulated by conditioning the receiver on a peer’s opinion, we need a way to pair their baseline opinions in a controlled and comparable manner. Concretely, for an agent’s set of outputs on an instance, we are interested in a fixed set of C label-explanation pairs for each available label. Thus, if an agent produces both labels for a given instance, we construct two separate opinion sets, one for each label. Figure 1b illustrates this construction.

The construction of opinion sets serves several purposes. First, it enables pairing of agents across all possible label configurations for a consistent number of repetitions, ensuring that the influence measure is grounded in the same number of interactions. Second, it allows us to examine all interaction configurations supported by the agents’ baseline outputs, including rare but valid label choices. Finally, it enables a more fitting alignment of outputs, since aligning outputs directly by repetition index would risk introducing spurious agreement or disagreement between agents driven purely by output ordering (see Appendix A for an example).

Formally, for each agent a^k and instance s_i , we

construct an opinion set of size C for each label available in O_i^k . The set is formed by fixing the label to y and sampling C explanations with replacement from the subset of baseline outputs where agent a^k predicted y :

$$\tilde{O}_i^k(y) = \{\tilde{o}_{i,c'}^k(y); c' = 1, \dots, C\}, \quad (4)$$

where each element $\tilde{o}_{i,c'}^k(y) = (y, \tilde{e}_{i,c'}^k(y))$ and $\tilde{e}_{i,c'}^k(y)$ is a sampled explanation from the baseline output.

2.3 Agent Interaction

The interaction between agents takes place when a receiver agent a^k is prompted to solve s_i again, where the receiver has access to its own opinion $\tilde{O}_{i,c}^k(y)$ and to the peer opinion $\tilde{O}_{i,c}^j(y')$. This produces a final label $\hat{y}_{i,c}^{k|j}$. The full round consists of the receiver being prompted all C repetitions of the constructed opinion set, with $\tilde{O}_i^k(y)$ fixed at y and $\tilde{O}_i^j(y')$ fixed at y' . Figure 1c illustrates this interaction process.

2.4 A Measure of Influence

Given a receiver with baseline certainty $p_i^k(y')$ and a peer suggesting label y' , *post-interaction certainty* is calculated as the proportion of target labels

y' :

$$p_i^{k|j}(y') = \frac{1}{C} \sum_{c=1}^C \mathbb{I}(\hat{y}_{i,c}^{k|j} = y'), \quad (5)$$

where $\hat{y}_{i,c}^{k|j}$ denotes the predicted label in repetition c after interaction with a peer agent.

We define the *directional change* toward the peer label y' as

$$\Delta_i^{k|j} = p_i^{k|j}(y') - p_i^k(y'). \quad (6)$$

This captures how the receiver's label distribution shift relative to its baseline certainty for a single instance.

This measure can either take a positive or negative value, where a positive value indicates a shift towards a peer's proposed label y' and a negative value indicates the opposite. We account for the receiver's baseline certainty of its opinion toward y' , to ensure that observed influence reflects a genuine shift caused by interaction with the peer, rather than a pre-existing tendency of the receiver.

The positive directional change is bounded by the receiver's available room for movement toward the peer's proposed label, y' . We call this bound the *maximum achievable change* and it is defined by:

$$\Omega_{+,i} = 1 - p_i^k(y'). \quad (7)$$

On the other hand, the negative directional influence score is bounded by the receiver's available room for shifting towards the opposite label of y' :

$$\Omega_{-,i} = 1 - p_i^k(1 - y'). \quad (8)$$

We provide an example in [Figure 2](#) that displays an example of a receiver that exhibits a directional change in the positive direction.

For an agent pair (a^k, a^j) , we aggregate directional changes across interactions by scaling the sum of directional change by the sum of all maximum achievable change. An aggregated positive influence score between the receiver a^k and peer a^j , for N instances, is then defined as:

$$IS_+^{k|j} = \frac{\sum_i^N \max(\Delta_i^{k|j}, 0)}{\sum_i^N \Omega_{+,i}}. \quad (9)$$

A negative influence score is defined as:

$$IS_-^{k|j} = \frac{\sum_i^N \min(\Delta_i^{k|j}, 0)}{\sum_i^N \Omega_{-,i}}. \quad (10)$$

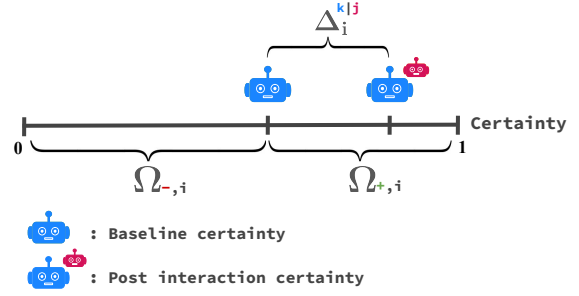


Figure 2: An illustration of the directional change for a single receiver-peer pair. The axis represents the receiver's certainty in the peer's proposed label y' , ranging from 0 to 1. The directional change $\Delta_i^{k|j}$ is the movement from baseline to post-interaction certainty. Positive movement corresponds to a shift toward y' bounded by the maximum achievable change $\Omega_{+,i}$, while any negative movement corresponds to a shift away from y' bounded by $\Omega_{-,i}$.

We normalise by the maximum achievable change in order to make influence scores comparable across instances. If the receiver already has a high baseline certainty of the proposed label y' , there is limited room for further movement toward that label. The normalised score therefore captures the realised change relative to the maximum change that was possible.

2.5 Match Types

We split possible interactions between two agents into categories we call *match types*. Match types describe the available opinion sets between a receiver and a peer for a given instance. Thus, it explains the possible label configurations in which they can interact. By defining match types, we can systematically map the full label configuration space and ensure that all relevant forms of agent interaction are represented.

We denote the match type between a receiver and peer agent for a given instance as $r:p$, where r and p refer to the available opinion sets of the receiver and the peer, respectively. If r or p is assigned 0 or 1, it means that the agent only possesses the opinion set $\tilde{O}_i^k(y=0)$ or $\tilde{O}_i^k(y=1)$, respectively, whereas B denotes that the agent has available opinion sets for both labels.

Further, match types are split into agreeing and disagreeing interactions. A disagreeing interaction is constructed when the receiver and peer have each produced opposing labels at least once in their baseline outputs for the same instance. Whereas an agreeing interaction is constructed when the

same label appears at least once in both the receiver’s and peer’s baseline outputs. Thus, the receiver and peer will represent the following opinion sets $\tilde{O}_i^k(y)$ and $\tilde{O}_i^j(y')$, respectively, where $y = y'$ defines an agreeing interaction and $y \neq y'$ a disagreeing interaction. E.g., the match type B:0 then encompasses both an agreeing and disagreeing interaction. In the agreeing interaction the opinion sets both hold label 0 and in the disagreeing interaction the receiver’s opinion set holds the label 1 and for the peer’s, 0.

The match types that can produce an *agreeing interaction* are the following: B:1, B:0, 0:B, 1:B, 1:1, 0:0 and B:B. And the match types of *disagreeing interactions* are the following: B:1, B:0, 0:B, 1:B, 1:0, 0:1 and B:B. Note that whenever two agents have a B:B match type on an instance, all possible label configurations are available.

Different match types may describe the same label configuration between pairs. For example, even though B:0 and 1:0 both describe a disagreeing interaction, they represent different starting points of the receiver agent. For B:0, the receiver has produced inconsistent labels in its baseline output, indicating lower certainty in label 1 prior to interaction. In contrast, 1:0 indicates that the receiver is fully committed to label 1 before interaction. We may expect the receiver to react with different levels of susceptibility to influence when interacting with a peer depending on its initial strength of belief. Likewise, r :B describes a peer that isn’t fully committed to the label in its baseline outputs, and may therefore exert less influence than a peer expressing full certainty. We therefore distinguish between these cases, as they represent different levels of baseline certainty.

2.5.1 Match-Type Aggregated Influence Score

For each receiver-peer pair, we compute both a negative and positive influence score per match type, and then average across match types, producing one final positive and one negative influence score. This ensures that the scores represent model behaviour across different influence directions and receiver starting points. Thus, avoiding any single match type inflating the score due to empirical frequency.

In total, the influence between a receiver and peer is described by four aggregated influence scores: a positive and negative influence score for disagreeing interactions, and a positive and negative for agreeing interactions. Throughout, we

mainly focus on the disagreeing configurations, however the same methods can be applied to the agreeing configurations.

Let $\mathcal{M}^{k,j}$ denote the set of match types available for receiver a^k and peer a^j , and let $IS_+^{k|j,m}$ and $IS_-^{k|j,m}$ denote the positive and negative influence scores computed for the instances within match type m . The final aggregated influence scores are then defined as

$$IS_+^{k|j,m} = \frac{1}{|\mathcal{M}_+^{k,j}|} \sum_{m \in \mathcal{M}_+^{k,j}} IS_+^{k|j,m} \quad (11)$$

and

$$IS_-^{k|j,m} = \frac{1}{|\mathcal{M}_-^{k,j}|} \sum_{m \in \mathcal{M}_-^{k,j}} IS_-^{k|j,m}. \quad (12)$$

From this point onward, we use the terms *positive influence score* and *negative influence score* to refer to these averaged scores.

The two scores [Equation 11](#) and [Equation 12](#) should be interpreted jointly. A positive influence score captures whether the peer has influenced the receiver towards its opinion and a negative score indicates whether the receiver shows resistance to the peer opinion. In a disagreeing configuration, a negative score can be attributed to a backfiring effect, where agents become reinforced in their opinions when confronted with contradictory information ([Cau et al., 2025](#)). However, in an agreeing configuration, a negative score can not be attributed to the same mechanism since the peer is reinforcing rather than challenging the receiver’s position. Any observed negative shift is instead likely a random unforeseen effect.

When analysing the interaction dynamics between a model pair, the influence scores describe both receiver and peer behaviour. From a receiver perspective, the influence scores describe its overall response to influence under interaction. Whereas, from the peer perspective, the same scores describe the response the peer induces when it is paired with a receiver. Thus, the two influence scores jointly describes how interactions between model pairs are characterised. Either the receiver tends to adopt the peer’s proposed label, remain stable under interaction, or react with resistance to the proposed label.

3 Experimental Setup

3.1 Datasets

We make use of three datasets covering three distinct but complementary NLU tasks: sarcasm detection, common sense reasoning and sentiment analysis. These datasets are intentionally selected to represent varying levels of signal strength, meaning the degree to which an instance contains information that is predictive of the target label. The lower the signal, the more the ambiguity we see in the instances. By covering both high- and low-signal settings, we can analyse how models behave under varying levels of ambiguity. In low-signal tasks, where inputs provide weaker evidence for the label, we expect greater disagreement between agents and lower confidence in their predictions, potentially leading to different influence dynamics compared to high-signal tasks.

Sarcasm Detection. Sourced from the SocKet repository by Khodak et al. (2017), this dataset consists of textual claims with a binary label indicating whether each claim is *Literal* or *Sarcastic*. We choose to include sarcasm detection, since it is inherently ambiguous and context-dependent, often giving challenges for even human annotators. As a result, it represents a low-signal task where the correct label is not immediately obvious, making it a challenging setting for agent interaction. We filter out low-quality instances, further described in Appendix B. The final dataset consists of 137,454 claims with 52% of the claims being Sarcastic.

Commonsense. CommonsenseQA 2.0 consists of questions paired with binary *Yes/No* labels (Talmor et al., 2022). Compared to sarcasm detection, this task exhibits a higher level of signal. While the questions are intentionally designed to be challenging, they are constructed and validated by humans to ensure clear answers. This makes the task less ambiguous than sarcasm detection, while still requiring the agents to rely on common sense reasoning. The dataset consists of 9,264 claims and has a fully balanced label distribution. No additional filtering is applied to this dataset, as it is already a well-curated benchmark.

Sentiment Analysis. Obtained from the SocKet repository (Khodak et al., 2017), this sentiment dataset consists of claims paired with a binary sentiment label, indicating whether the claim has a *Positive* or a *Negative* sentiment. This dataset rep-

resents the highest level of signal among the three tasks. As a result, the ambiguity in the ground truth labels is more limited. This setting therefore allows us to study agent interactions in scenarios where errors are more likely to stem from limitation in the models, rather than from ambiguity in the data itself. Pre-processing steps are described in Appendix B. The final dataset consists of 91,964 claims with 54% of the claims being Positive.

3.2 Models

We evaluate a diverse set of well-known open-source models spanning different parameter sizes, developer organisations, and countries of origin. We report the country of origin, developer and model information (model-version-size):

- (US) **Google** : gemma-3-4b, gemma-3-27b
- (US) **Meta** : llama-3.1-8b, llama-3.3-70b
- (CN) **Alibaba Cloud** : qwen-2.5-7b, qwen-2.5-72b
- (US) **OpenAI** : gpt-oss-20b

For brevity, we omit version identifiers, and will refer to the models by the following naming-convention: model-size.

We use 4-bit quantised versions (bitsandbytes) of all models. We use the default sampling parameters specified by the authors, provided in the generation configuration file from their Hugging Face repository. If no generation configuration files are specified, parameters are set to values that has been reported as suitable by other sources. We refer to Appendix C for additional information about the models and their parameters.

3.3 Homogeneous and Heterogeneous Configurations

We evaluate both homogeneous and heterogeneous system compositions. In the homogeneous setting, the agent pair consists of agents of the same model type. Here the peer opinions presented to the receiver are generated by the same underlying model. In this system composition, a disagreeing pairing of outputs can only be constructed for instances where the model has produced mixed labels in its baseline predictions. This is due to the fact, that both the receiver and the peer rely on same baseline output, and thus a disagreeing interaction on that instance is not possible if both labels are not available.

In the heterogeneous setting, we consider all pairwise configurations of the largest parameter size model in each model family. Additionally, we examine intra-family configurations, by pairing large and small models within the same model family, allowing us to study the effect of intra-family influence dynamics.

3.4 Prompts

We use a simple and neutral user prompt structure for both rounds, keeping the format consistent. All prompts follow a similar system prompt, which defines an output structure for the given dataset and round.

In the first round, the model is presented with the task instruction and an instance, and is asked to provide a label along with a brief explanation of its reasoning. In the second round the prompt is similar, but additionally conditions the model on its own baseline opinion, and the peer agent’s baseline opinion. This context is presented in a neutral manner, where its own opinion is presented with "You previously said:" and the peer’s opinion as "However, a peer of yours said:". The model is then asked to re-evaluate its label conditioned on this input, providing only a label this time. The user and system prompts are provided in [Appendix D](#).

3.5 Additional Parameters

For all experiments, we set $C = 10$. To reduce computation we introduce a capacity on the number of instances per match type chosen for the interaction round. Specifically, we introduce a capacity of 7,000 instances (70,000 prompts) to each disagreeing match type, and a capacity of 1,000 instances (10,000 prompts) to each agreeing match type. The smaller capacity on the agreeing cases is chosen since these are less interesting configurations, as models are expected to remain stable when exposed to a confirming peer opinion. For the agreeing cases, we further restrict the interaction instances to the 0:0 and 1:1 match types, i.e., cases where both agents have full baseline certainty.

4 Results

4.1 Baseline Behaviour

We report the distribution of labels based on a majority vote of the 10 repetitions for each instance, with a focus on the positive label (see [Table 1](#)). An instance is labelled positive if at least 50% of the repetitions are to the positive label. A small

number of instances produced outputs with invalid formatting and were therefore discarded from the baseline analysis and excluded from all subsequent pairwise interaction (see [Appendix E](#)).

MODEL	SARCASM Sarcastic (%)	COMMONSENSE Yes (%)	SENTIMENT Positive (%)
gemma-27b	92.79	49.19	46.96
gemma-4b	96.39	39.23	43.12
llama-70b	77.49	77.49	51.85
llama-8b	97.63	44.76	57.00
qwen-72b	85.27	41.39	46.58
qwen-7b	98.42	20.85	49.40
gpt-20b	49.65	67.18	46.90

Table 1: Positive rate based on the majority vote across the 10 baseline predictions for each instance. Reported in %.

Despite the balanced ground truth label distribution of the sarcasm dataset, all models except gpt-20b show a strong tendency to over predict the positive label. We hypothesise that this is due to the low signal strength of the instances, causing the models to collapse toward the sarcastic label. The commonsense and sentiment datasets generate more balanced prediction distributions, with a few exceptions, namely qwen-7b and llama-70b on commonsense.

We also report the proportion of claims for which a model produces consistent versus inconsistent predictions across the 10 repetitions (see [Table 2](#)). An instance is considered consistent if the model assigns the same label across all repetitions, corresponding to a baseline certainty of 1 for that label, and inconsistent otherwise, where the baseline certainty falls below 1. This provides an indication of each model’s overall decisiveness, where a high proportion of consistent predictions suggests that the model is consistently confident in its baseline opinions.

Across models and datasets, the majority of claims receive consistent predictions. Gpt-20b is the only outlier, showing by far the highest rate of inconsistent predictions across all datasets, most prominently on sarcasm with an inconsistent label rate of 58.09%, compared to 14.21% for the second highest model. Besides gpt-20b, models are largely consistent in their predictions, with llama-8b as the only exception having an inconsistent label rate of 43.46% on the commonsense dataset, while all other inconsistent label rates stay below 14.21%.

MODEL	SARCASM			COMMONSENSE			SENTIMENT		
	Pos. (%)	Neg. (%)	Pos./Neg. (%)	Pos. (%)	Neg. (%)	Pos./Neg. (%)	Pos. (%)	Neg. (%)	Pos./Neg. (%)
gemma-27b	91.61	6.33	2.05	47.93	49.47	2.60	46.19	52.36	1.45
gemma-4b	95.88	3.25	0.87	36.68	58.01	5.32	42.38	56.30	1.32
llama-70b	73.31	19.20	7.50	43.04	53.55	3.43	51.03	47.53	1.44
llama-8b	85.51	0.29	14.21	43.32	13.22	43.46	48.18	42.85	8.98
qwen-72b	82.12	12.83	5.05	38.92	56.31	4.77	45.56	52.70	1.73
qwen-7b	98.17	1.39	0.44	19.25	77.57	3.18	48.21	49.91	1.97
gpt-20b	18.78	23.13	58.09	19.14	45.13	35.73	37.55	46.13	16.32

Table 2: Distribution of label consistency per instance. *Pos.* and *Neg.* denote the percentage of instances a model has assigned exclusively positive or negative labels, respectively. *Pos./Neg.* denotes the percentage of instances with inconsistent labels, and thus reports the inconsistent label rate.

4.2 Interaction Dynamics

We begin by examining the overall influence dynamics in disagreeing interactions, where the peer and receiver are interacting with opposing opinion sets. Figure 3 shows the positive and negative influence scores for all large models.

Substantial influence under disagreement.

Across all three datasets, positive influence between models is widespread. Many model pairs achieve influence scores above 0.5, indicating that the peer is often able to shift the receiver towards its proposed label when the two agents disagree. This pattern is present across all three datasets. In the lowest signal task, sarcasm, we see high variance between models’ overall susceptibility to influence. As the task signal becomes stronger, this variation becomes less clear. Susceptibility varies less between receiver models in the commonsense dataset, and is further reduced in sentiment, where the task signal is strongest.

Models with consistently high baseline certainties are the most susceptible to influence.

Interestingly, the receivers with the highest marginal susceptibility scores are also those with the highest overall rate of consistent baseline predictions. For example, gemma-27b in the sarcasm dataset produces consistent labels on 99.95% of instances, yet achieves an average positive influence score of 0.89, indicating that it is among the most susceptible receivers despite rarely expressing uncertainty in isolation. This pattern is reflected in a positive Spearman’s rank correlation between the proportion of consistent labels and the marginal receiver-level positive influence scores ($\rho = 0.61$, $p = 0.036$) across all large model and dataset combinations. This is unexpected, since decisiveness might be expected to indicate robustness. Instead,

the results suggest that models that appear decisive in isolation do not necessarily translate to stability under interaction.

Influence is directional across model pairs.

The results in Figure 3 show that influence depends strongly on which model acts as the receiver and which model acts as a peer. A model may exert strong influence as a peer while still being highly susceptible as a receiver. In this sense, the influence dynamics are directional. The ability to influence others and the tendency to be influenced are therefore two separate properties. Moreover, reversing the peer and receiver roles does not necessarily produce the same effect.

Gemma-27b is the most susceptible receiver across all three datasets with receiver marginals ranging from 0.78 to 0.89, and near-zero negative influence scores throughout. Influence on gemma-27b is not only frequent but tends to be near-complete shifts towards the peer’s opinion. This can be seen in Figure 4 where a majority of its post-interaction certainty is pushed close to 1.0 across the full range of baseline probabilities. Despite the model’s high susceptibility, it is simultaneously an influential peer, producing consistently high positive influence scores across all receivers and datasets.

Gpt-20b stands out as the only model that exerts high influence while also remaining stable under influence. This is most pronounced in sarcasm, where gpt-20b achieves positive influence scores ranging from 0.53 to 0.89, and has the lowest marginal susceptibility of 0.28. As a receiver, gpt-20b generally has negative influence scores close to zero across datasets, with the exception of sarcasm, where it reaches a marginal negative influence score of -0.23 .

Llama-70b shows similar behaviour to gpt-20b

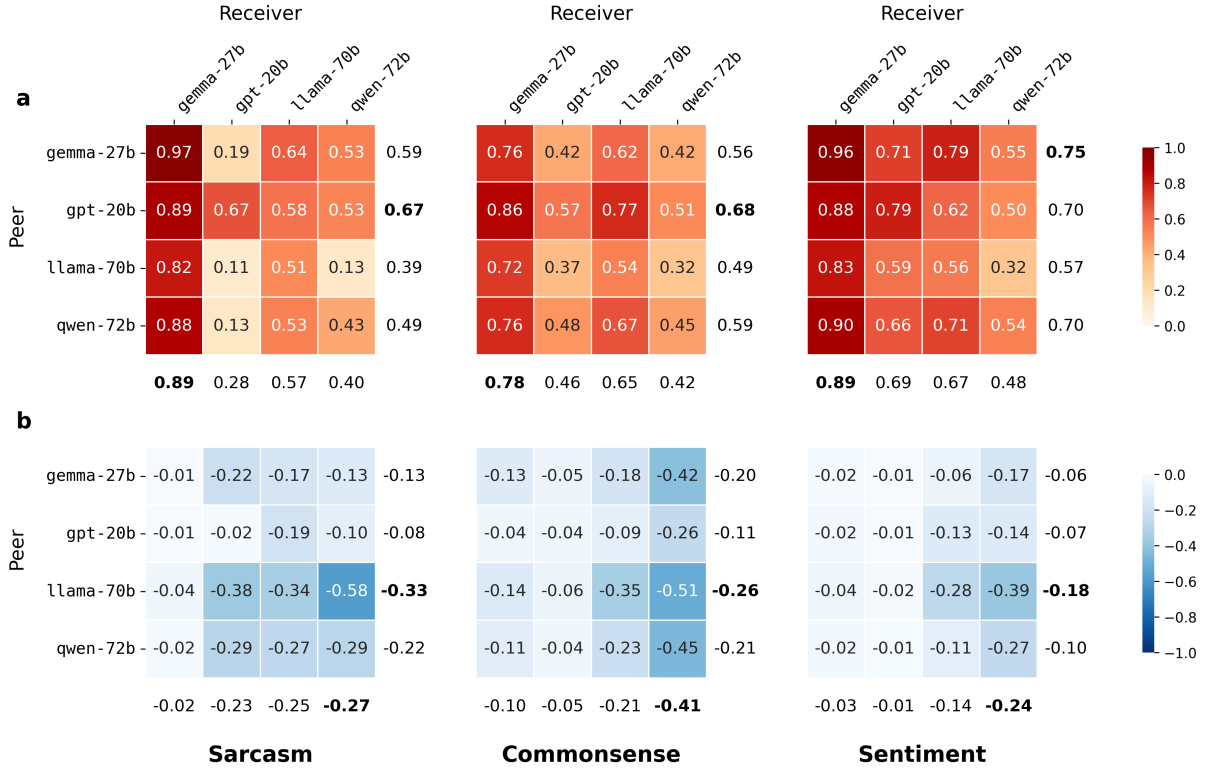


Figure 3: Positive and negative influence scores across model pairs and datasets. All values are calculated from disagreeing interactions. For the positive influence scores (a), the columns show how susceptible a receiver is to influence from the different peers, while the rows show how much influence a peer can exert on different receivers. Whereas for the negative influence scores (b), the columns represent how much a receiver backfires in reaction to peer influence, and the rows show how much a peer can induce backfiring in a receiver. The marginals report the average influence scores, with the largest absolute marginal in each row and column highlighted in bold.

but with slightly higher receiver marginals and lower influential power. The negative influence scores however, show a stronger backfiring effect.

Qwen-72b shows a more ambiguous case. Examining only the positive influence scores in Figure 3a, the model initially appears resistant, with marginals consistently below those of gemma-27b and llama-70b. However, this interpretation changes substantially when the negative influence scores in Figure 3b are considered. Here, qwen-72b is the receiver having the strongest backfiring effect across all three datasets, with marginals of -0.27 , -0.41 , and -0.24 for the sarcasm, commonsense and sentiment dataset, respectively. A lot of the influence that is not expressed as movement toward the peer opinion, appears as movement in the opposite direction instead of stability. This behaviour is especially prominent when interacting with llama-70b (see Figure 4).

Response to peer interaction differs across receivers. Gemma-27b and qwen-72b both exhibit strong response to peer interaction but this un-

folds in fundamentally different ways, which is especially pronounced in the sarcasm highlighted by Figure 4. For gemma-27b this response mainly takes the form of consistent adoption, since we see a majority of conditional probabilities concentrated in the upper row of the heat maps. Whereas for qwen-72b peer disagreement tends to induce either strong adoption or strong resistance, rather than moderate adjustment as observed for gpt-20b. Plots of the conditional distribution of post-interaction certainty for the two remaining datasets show the same patterns and are reported in Appendix F.

4.2.1 Agreeing Interactions

We also examine agreeing interactions on the sarcasm dataset, where the peer and receiver share the same starting opinion. Across models, these interactions show stable behaviour with near-zero negative influence scores. These results are as expected, and all scores are reported in Appendix G.

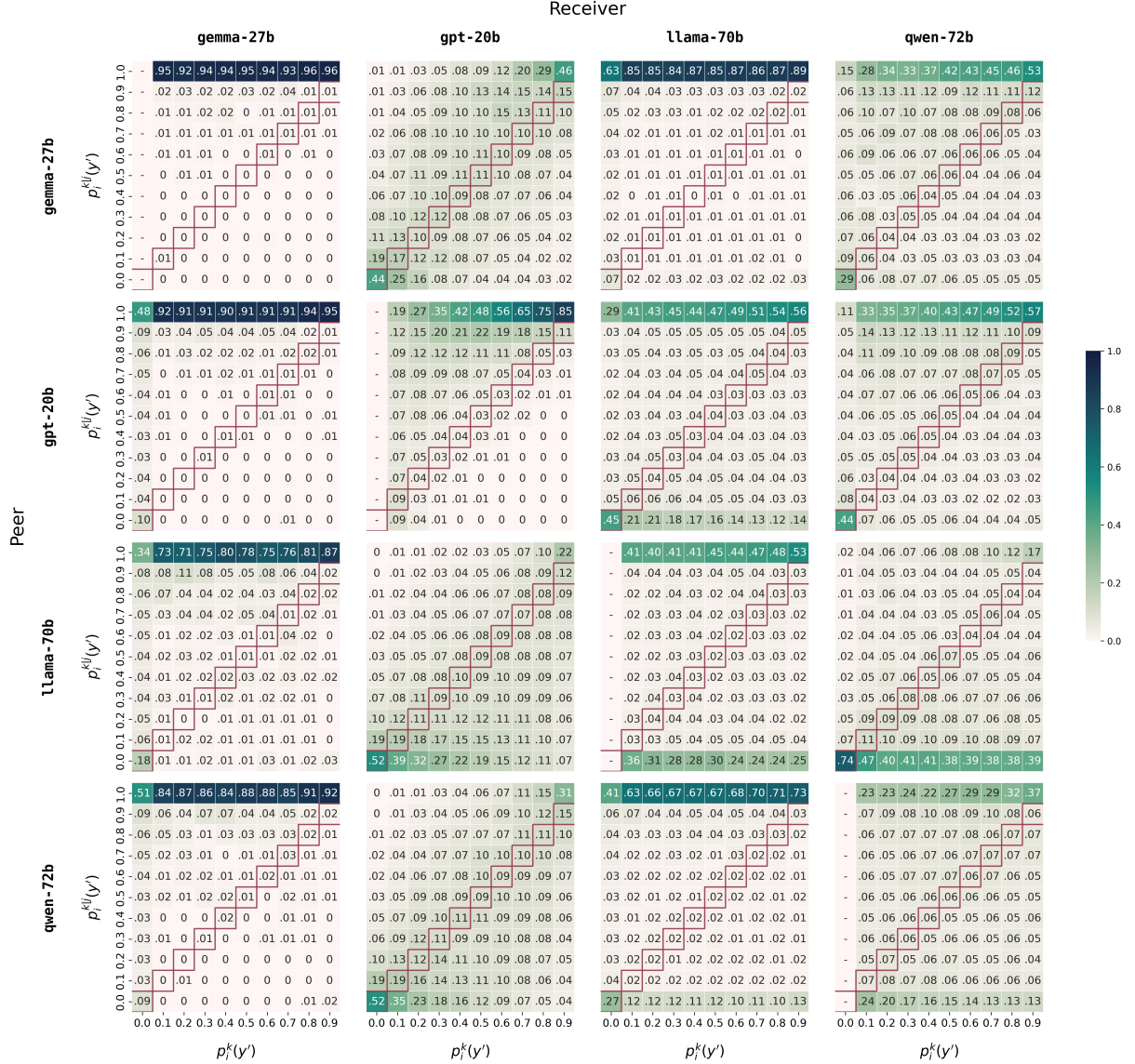


Figure 4: Conditional distribution of post-interaction certainty given baseline certainty for each receiver-peer pair on the sarcasm dataset. A concentration of probability mass on the diagonal (marked with red) indicates that the receiver’s post-interaction certainty mirrors its baseline, suggesting stability under influence. Mass concentrated above the diagonal reflects positive influence, where mass below the diagonal are observations of negative influence. Plots on the diagonal correspond to homogeneous systems. The leftmost column in these plots is empty since only B:B match types are possible and thus excluding any instances where the baseline certainty is 0 or 1.

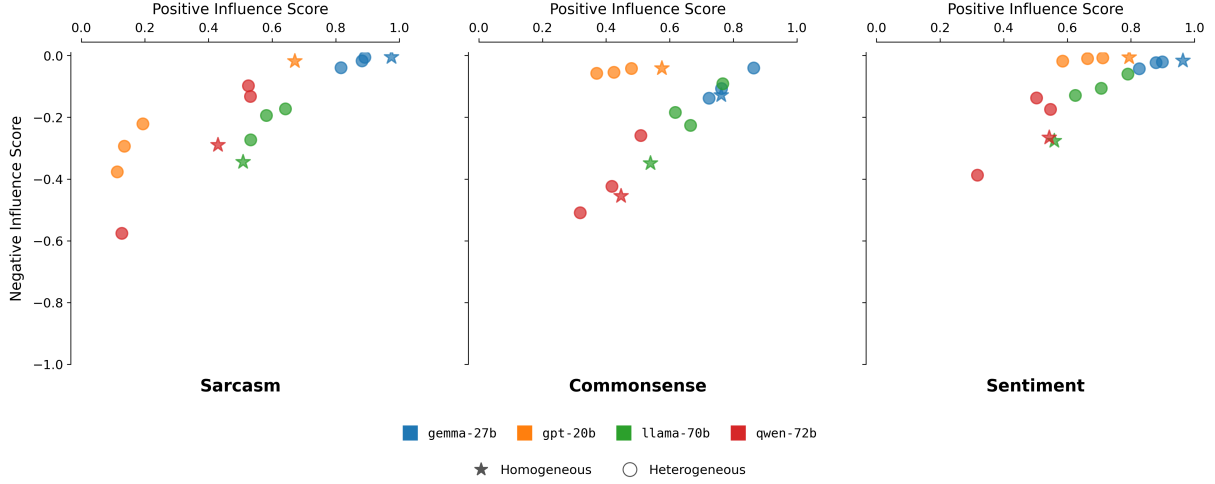


Figure 5: Influence patterns coloured by receiver, with stars denoting homogeneous interactions and circles denoting heterogeneous interactions. Positive influence is shown along the x-axis, and the negative influence is shown along the y-axis. A robust receiver will be placed close to the origin. If a receiver lies in the upper-right region, it is highly influenced towards the peer’s label, whereas if it lies in the lower-left region, it shows strong backfiring tendencies. A separation between stars and circles of the same colour indicates that the receiver behaves differently when the peer information comes from itself rather than from another model.

4.3 Homogeneous vs. Heterogeneous Interaction

We next compare homogeneous and heterogeneous interaction dynamics in Figure 5. The figure shows that the difference between the two system compositions is strongly receiver-dependent. Rather than observing one general pattern across all models, each receiver exhibits a distinct susceptibility profile, meaning that the effect of system composition depends on which model is being influenced. This is reflected in the separation between the homogeneous and the heterogeneous systems in the figure. The clearest differences are observed for llama-70b and gpt-20b, which both show a consistent but different pattern across all three datasets. We further investigate the transition from homogeneous to heterogeneous by reporting the percentage differences between an average of receiver susceptibility in heterogeneous systems compared to its homogeneous system. Here, we discover that llama-70b becomes more susceptible to influence, when transitioning from homogeneous to heterogeneous, whereas gpt-20b becomes less susceptible. Llama-70b increases in susceptibility by 13%, 27% and 26% on sarcasm, commonsense and sentiment respectively. In contrast, gpt-20b decreases in susceptibility by 79%, 25% and 17% across the same datasets. All percentage differences between homogeneous and heterogeneous systems are reported in Appendix H.

For the remaining models, the homogeneous and heterogeneous settings show less decisive separation. Moreover, when the same comparison is made from the peer perspective, no similarly consistent patterns can be seen (see Appendix I). Overall, these results suggest that heterogeneity does not uniformly increase or decrease susceptibility to influence, but rather changes the influence dynamics in a receiver-specific way.

4.4 Sensitivity Analyses

We perform four sensitivity analyses to examine the effect of isolated design choices on the influence scores reported above. Three of the analyses concern the prompt-level variations in terms of context and one examines the effect of the peer’s sampling temperature.

All experiments are conducted on the sarcasm dataset, for the model pair qwen-72b and llama-70b, unless otherwise stated. This pair is selected due to their comparable parameter size and sampling parameters. Furthermore, this pair exhibited asymmetry in their influence scores in the main experiment, making them a particularly informative case for examining the stability of influence dynamics. The analyses are limited to a single model pair due to computational constraints. The results are reported in Table 3. The table shows the resulting influence scores of the sensitivity experiments alongside the main experiment, for the

EXPERIMENT	RECEIVER $\leftarrow$ PEER	$IS_+^{k j}$			$IS_-^{k j}$		
		MAIN	SENSITIVITY	Δ	MAIN	SENSITIVITY	Δ
<i>Temperature</i>	llama-70b $\leftarrow$ qwen-72b-high	0.53	0.53	0.00	-0.27	-0.27	0.00
	llama-70b $\leftarrow$ qwen-72b-low	0.53	0.52	-0.01	-0.27	-0.29	-0.02
	qwen-72b $\leftarrow$ llama-70b-high	0.13	0.12	-0.01	-0.58	-0.57	+0.01
	qwen-72b $\leftarrow$ llama-70b-low	0.13	0.13	0.00	-0.58	-0.57	+0.01
<i>Memoryless Receiver</i>	llama-70b $\leftarrow$ qwen-72b	0.53	0.53	0.00	-0.27	-0.23	+0.04
	qwen-72b $\leftarrow$ llama-70b	0.13	0.76	+0.63	-0.58	-0.03	+0.55
<i>No Explanation</i>	llama-70b $\leftarrow$ qwen-72b	0.53	0.45	-0.08	-0.27	-0.26	+0.01
	qwen-72b $\leftarrow$ llama-70b	0.13	0.53	+0.40	-0.58	-0.09	+0.49
<i>Swap</i>	llama-70b $\leftarrow$ gpt-20b	0.58	0.43	-0.15	-0.19	-0.38	-0.19
	gpt-20b $\leftarrow$ llama-70b	0.11	0.39	+0.28	-0.38	-0.08	+0.30

Table 3: Positive and negative influence scores, comparing results across all sensitivity analyses to the corresponding main experiment. $\Delta = \text{main} - \text{sensitivity}$. A positive Δ in $IS_+^{k|j}$ indicates increased positive influence, while a negative Δ in $IS_-^{k|j}$ indicates increased backfiring.

given model configuration.

4.4.1 Temperature

We examine whether varying the peer’s sampling temperature affects receiver behaviour. Prior work suggests that, for problem-solving multiple-choice question answering tasks, temperatures between 0.0 and 1.0 typically do not significantly affect accuracy, while performance can degrade beyond this range (Renze, 2024). We therefore vary the peer temperature across $t \in \{0.2, \text{model default}, 1.2\}$, where 1.2 is included to test whether a higher-variation setting, associated with more diverse and creative outputs, changes the peer’s ability to induce influence (Li et al., 2025).

The resulting influence scores in Table 3 show little to no differences across temperature configurations, when compared to the results of the main experiment. This suggests that peer temperature has little effect on receiver susceptibility in either direction. This result is limited to the specific setting considered here, and does not rule out that other sampling parameters, or changes to the receiver’s temperature, could affect the observed influence scores.

4.4.2 Memoryless Receiver

We remove the receiver’s own prior opinion from the prompt, such that the receiver is conditioned solely on the peer’s opinion when re-evaluating its label. We investigate the effect of the receiver’s own opinion, included in the prompt, on the observed influence scores. The details of the prompt can be seen in Appendix J.

The two models respond quite differently to this change. Llama-70b shows almost no fluctuations,

with only a small decrease in backfiring compared to the main experiment. Qwen-72b however, shows a substantial increase in positive influence alongside a corresponding decrease in backfiring. This suggests that inclusion of its own opinion, acts as a support that mitigates susceptibility to peer influence and without it, susceptibility increases.

4.4.3 No Explanations

We remove the explanations from both the peer and receiver opinions, keeping only the labels when prompting the agent to reevaluate its decision. We do this in order to examine the effect of both peer and receiver explanations. The details of the prompt can be seen in Appendix K.

Llama-70b, again, remains stable relative to the main experiment. Qwen-72b shows different behaviour, again, with a higher positive influence score and a reduced backfiring effect, consistent with the pattern observed under the memoryless condition. Again, the inclusion of reasoning of a label is a contributing factor to model stability, and the exclusion of such may result in a more vulnerable model.

4.4.4 Swap

In this experiment, the prompt structure is changed such that the receiver’s own opinion is presented as a peer opinion, and the peer’s opinion is presented as the receiver’s own opinion (see Appendix L for example of prompt). The aim is to examine whether this swap in role and opinion affects the degree of influence observed. Since influence is measured relative to the label the peer proposes, and the labels are now swapped, any movement toward the original peer opinion will appear as

negative influence. If the influence scores remain largely unchanged after the swap, it implies that the role from which an opinion is presented, likely is the primary driver of influence rather than the content of the explanation. Conversely, any change in scores quantifies the relative quality of the two agents’ reasoning. An increase in positive influence suggests the receiver finds its own prior explanations more persuasive than the peer’s, while a decrease suggests the opposite. Theoretically, if influence is driven entirely by argument quality, we would expect a full inversion with all positive influence from the main experiment becoming negative and vice versa.

In this experiment we focus on the influence dynamics between gpt-20b and llama-70b. The two models respond differently to the swap of roles from which opinions are presented. When llama-70b acts as the receiver and is presented with gpt-20b’s label and explanation as its own, and its own opinion is framed as the peer’s, the positive influence score decreases with 0.15. Furthermore, its backfiring effect strengthens by 0.19, when compared to the main experiment. The opposite effect is observed when gpt-20b acts as the receiver. Here, its positive influence score increases by 0.28 and its negative influence score decreases by 0.29.

These results suggest that the content of the explanation plays a role in driving influence. When llama-70b is exposed to its own reasoning as a peer opinion, it becomes less susceptible, suggesting its own explanations are less compelling as external input. Moreover, gpt-20b becomes more susceptible when its own reasoning is proposed to itself, where llama-70b’s explanations were less effective in the main experiment. Both models appear to assign greater weight to gpt-20b’s label-explanation pairs regardless of the role from which they are presented, suggesting that gpt-20b produces more persuasive reasoning than llama-70b. A comparative analysis of the two models’ explanations is provided in [Appendix M](#).

4.5 Intra-Family Influence Across Model Sizes

As a complementary analysis to the main experiment, we examine intra-family interactions across model sizes. Here, model family is kept fixed and the main source of heterogeneity becomes model size. [Figure 6](#) and [7](#) show several patterns that are consistent with the findings discussed in [Subsection 4.3](#). Gemma-27b again stands out as a high

influence model in both directions, producing large positive influence scores both as a peer and receiver. The model has marginal influence scores above 0.6, alongside near-zero negative influence scores.

Within the qwen family, positive influence dynamics follow the expected size hierarchy in both directions. Qwen-72b is consistently most resistant to influence and is a stronger peer, when compared to qwen-7b, across all datasets. A similar, but less prominent pattern also appears within the llama-family, where llama-70b consistently outperforms llama-8b as a peer across most receiver configurations. However, unlike qwen-72b the size advantage does not extend to resistance as a receiver, as llama-70b and gemma-27b tend to be more susceptible to positive influence than their respective smaller counterparts.

Considering the negative influence scores, the qwen family again follows the expected size hierarchy. The larger qwen-72b shows greater shifts away from the proposed label, while interactions with the smaller qwen-7b tend to induce more backfiring in receivers. Within the gemma and llama families, the pattern partially reverses, here the smaller models show greater shifts away from the proposed label, yet similarly to qwen-7b, they also induce more backfiring in others when acting as peers. This suggests that less authoritative peers tend to trigger resistance in receivers rather than persuasion toward the proposed label. This is consistent with the positive influence findings, where the smaller models also were the weaker peers. Together, these results suggest that interacting with a model of smaller parameter size, not only produce weaker positive influence, but may reinforce the receiver’s existing opinion.

4.5.1 Agreeing Interactions

Intra-family interactions, where models initially agree, largely show stable behaviour with near-zero negative influence scores on the sarcasm dataset. However, llama-8b stands out as a receiver with scores of -0.34 and -0.42 when interacting with itself and llama-70b, respectively. This behaviour is unexpected and possibly related to the model’s strong baseline toward the sarcastic label (see [Table 1](#)). Examining the scores at the match type level reveals that the effect is primarily driven by the 0:0 interactions, where llama-8b frequently shifts toward label 1 (see [Appendix N](#)).

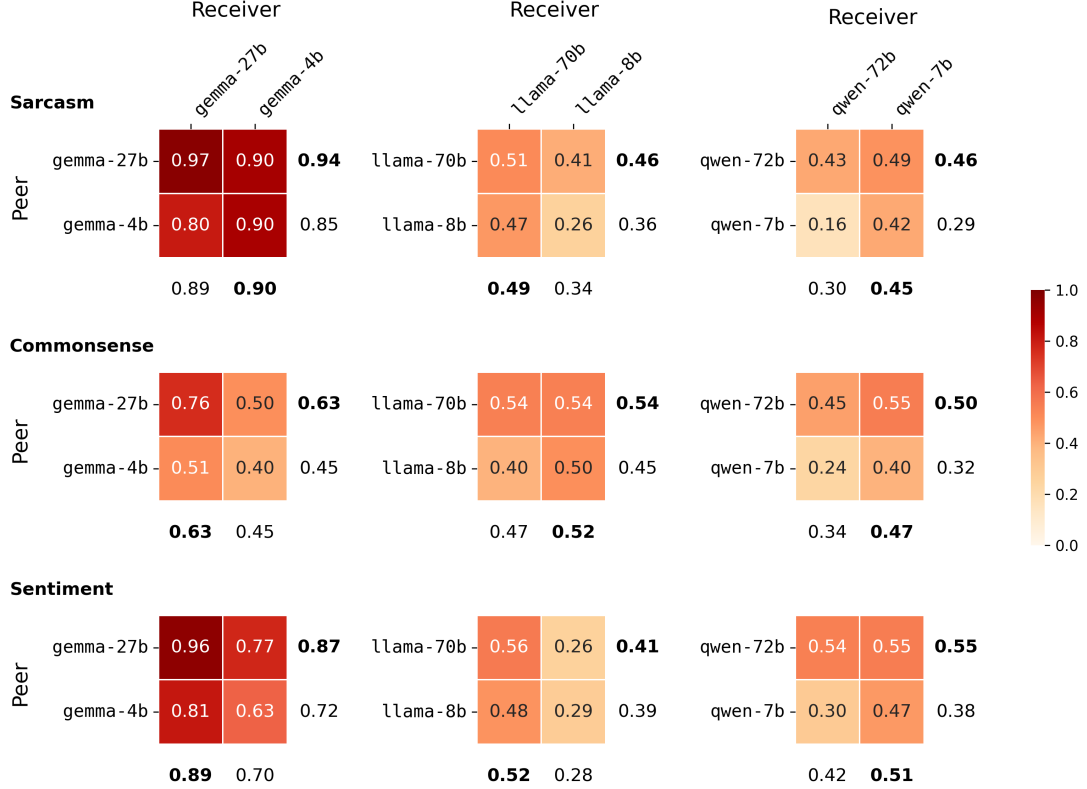


Figure 6: Positive influence scores on disagreeing intra-family pairs. The columns show how susceptible a receiver is to influence from either itself or its counterpart. The rows show how much influence a peer can exert on either itself or its counterpart. The marginals report the average influence scores, with the largest absolute marginal in each row and column highlighted in bold.

5 Discussion

Our findings reveal widespread influence in disagreeing interactions, with the dynamics being heavily dependent on the specific models involved. A central implication of these findings is that baseline behaviour alone gives an incomplete picture of how a model will behave in a multi-agent setting. In particular, our results show that high baseline certainty does not necessarily imply robustness to peer interaction. This is unexpected, since one would assume that a model with a confident baseline opinion would be less likely to change its answer when exposed to disagreement, compared to a model with a lower certainty.

A relevant question is not only whether a model is influenced, but also if the influence dynamics changes depending on the system type. Some models behave differently when interacting in a homogeneous setting compared to interacting with a different model type in a heterogeneous setting. For llama-70b, heterogeneous interaction appears to make it more susceptible to influence than the

homogeneous setting, and gpt-20b shows the opposite pattern. This suggests that there is no general guarantee that either homogeneous or heterogeneous MAS are more stable. In practical MAS design, this means that model diversity may improve system capabilities, but also introduce new unexpected influence dynamics.

Furthermore, a model’s level of susceptibility appears to be a consistent characteristic of the model. We see that highly susceptible models shift toward the proposed position regardless of which model type act as the peer. This is particularly relevant for MAS used in debate, where interaction is often assumed to improve reliability through critique and correction. If certain agents are systematically steered by others, these same interaction mechanisms may also be exploited to push the system toward a potential malicious conclusion. In a setting, where an adversarial aims to influence system outcomes in a particular direction, the effectiveness of the attack depends more on the susceptibility of the models in that system, than the influence power of the adversarial. This suggests that influence in

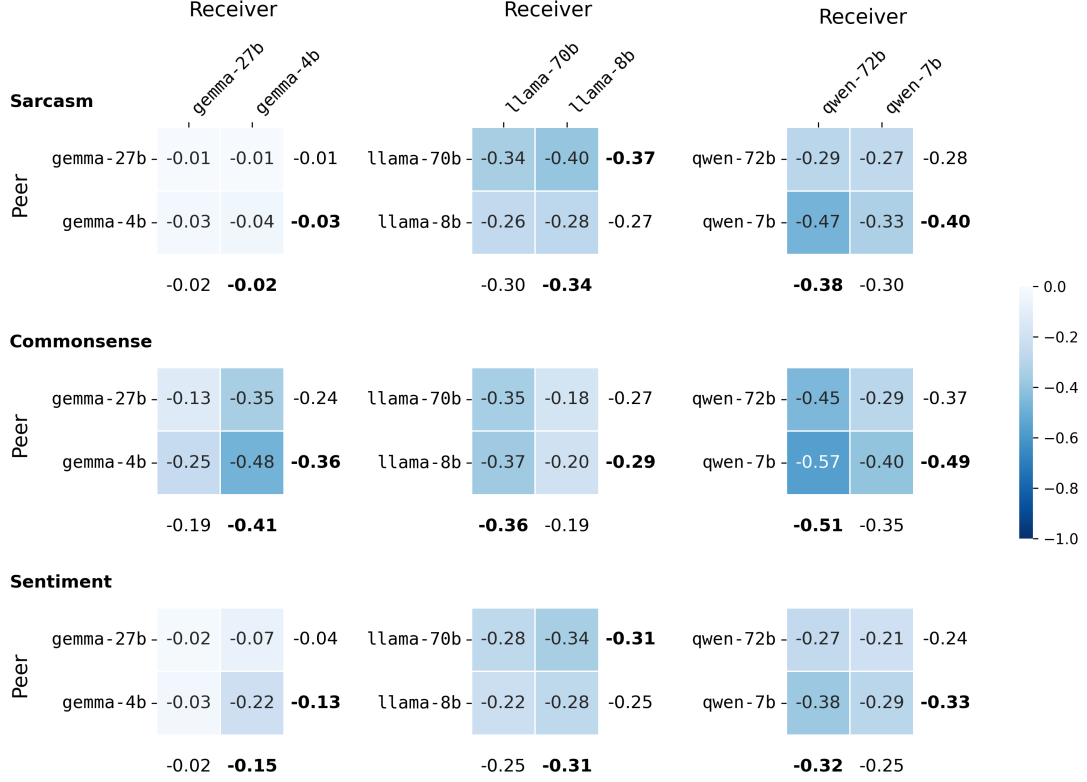


Figure 7: Negative influence scores on disagreeing intra-family pairs. The columns represent how much a receiver backfires in reaction to peer influence from either itself or its counterpart. The rows show how much a peer can induce backfiring in itself or its counterpart. The marginals report the average influence scores, with the largest absolute marginal in each row and column highlighted in bold.

heterogeneous MAS is not simply a property of model strength, but depends on the specific role a model plays in the interaction. Task signal strength also appears to be meaningful when considering potential malicious intent. Across our three datasets, we observe greater variance in influence scores on lower-signal tasks, where the correct label is more ambiguous. When the task provides weaker signal, models tend to vary more in how susceptible they are to peer influence. Model selection should therefore be treated as a signal-dependent decision. Lower-signal tasks demand more careful system design, both in terms of which models are included and what roles they play.

6 Related Work

Homogeneous Multi-Agent Systems. Much of the existing research on LLM-based multi-agent systems studies agents initialised by the same underlying model. In these homogeneous settings, emergent collective behaviour has been shown to arise, which is not observable in the individual models. For example, it has been shown that col-

lective biases can emerge during agent interaction that are undetectable when evaluating models in isolation (Coppolillo et al., 2025; Ashery et al., 2025). Agents have also been shown to display patterns of conformity to peer opinions. This tendency is strengthened when the gap between an agent’s self-confidence and perceived confidence of its peer, is increased (Cho et al., 2025).

Heterogeneous Multi-Agent Systems. The shift from homogeneous to heterogeneous MAS is motivated by the introduction of diverse knowledge from distinct models of varying capabilities. Empirical results suggest that heterogeneous systems consistently improve performance over homogeneous systems, as well as small groups of diverse agents sometimes exceeding much larger homogeneous systems in performance (Yang et al., 2026; Ye et al., 2025).

However, model diversity does not always guarantee better outcomes. Models that perform well individually can degrade substantially when required to collaborate, and heterogeneous interaction is also shown to be sensitive to the order in which

agents communicate (Davidson et al., 2025). Similarly, multi-agent debate can degrade performance and agents may shift their answers from correct to incorrect in response to peer reasoning (Wynn et al., 2025). In negotiations settings, stronger models sometimes lose to weaker opponents, further supporting that individual behaviour is not a sufficient indicator of model strength during interaction (Davidson et al., 2024).

Opinion Dynamics in Heterogeneous MAS.

There is an increasing focus on work that examines how LLM agents influence one another’s opinions during interaction. Susceptibility across models in MAS has been shown to depend both on the model type and the direction of the shift (affirmative $\rightleftharpoons$ negative) (Mehdizadeh and Hilbert, 2025). Similarly, flawed reasoning can play a big role in opinion change, and less agreeable inputs can even produce a backfiring effect (Cau et al., 2025). Other research finds that persuasion and resistance may be dependent properties and enhancing an agent’s ability to persuade triggers greater resistance toward external inputs in a system (Zhao et al., 2025). Bozdag et al. (2026) propose a framework for evaluating persuasiveness and susceptibility to persuasion in multi-agent interactions. Here they find clear differences between persuasiveness and susceptibility across models, where larger models demonstrate superior persuasive abilities.

Most research related to our work often studies opinion dynamics by involving agents explicitly tasked with providing persuasive arguments aimed to convince the counterpart. Rather than assigning agents persuasive or adversarial roles, we present peer outputs as neutral arguments in a binary classification task. This structure aims to isolate influence in a collaborative environment that arises from the interaction itself, rather than from deliberate argumentative effort.

7 Limitations

While this study provides insights into influence dynamics in heterogeneous MAS, it is important to acknowledge its limitations and the generalisability of our conclusions.

First, our framework considers agent interaction as single-turn and one-directional, evaluated on binary classification tasks. By constraining the interaction to a single turn, our results capture only the immediate effect, and may not reflect the more complex dynamics that may emerge across

repeated exchanges. While this controlled setting is well-suited for initial systematic exploration, it constrains the applicability of our results. By this, our findings are most applicable to polarised issues and short debates. This may not fully reflect dynamics in more complex environments with open-ended tasks.

Additionally, by constraining the system to exactly two agents, we are able to isolate the origin of influence. However, real-world MAS typically involve larger populations, where external information is presented by several different peers. As a result, our findings are mostly applicable to bi-agent systems, and whether the observed dynamics extend to larger population sizes can be further explored.

Furthermore, our results are based on a subset of open-source models. Despite efforts to include as much variety as possible (within the same parameter size range), our findings are not guaranteed to generalise to other LLMs. As demonstrated in this paper, influence dynamics between a given model pair can take markedly different forms depending on the models involved and the role they take. Whether closed-source or other open-source models exhibit similar behavioural patterns therefore remains an open question.

8 Conclusion

Our results show that interaction between LLM-based agents can substantially change model behaviour, even in highly confident models. Across three different datasets, we observe widespread influence in disagreeing interactions. This shows that agents exposed to external input often change their opinion. Our findings show that influence is directional and model dependent. A model can be highly influential while also being highly susceptible to influence, and switching the agents’ roles does not necessarily produce reverse effects.

A key finding is that baseline certainty does not ensure robustness against influence. Models that produce the largest proportion of consistent baseline predictions are not stable when exposed to peer opinions. This challenges the intuitive assumption that confident models are more stable in collaborative settings.

Task signal strength also impact influence. The low-signal task show greater variance in susceptibility across receivers, while for the high-signal task then influence patterns become more uniform.

This suggests that ambiguous tasks amplify the effect of model choice and interaction design. When transitioning from a homogeneous system to a heterogeneous system, results reveal a change in the influence dynamics that is receiver-specific. Some models become consistently more robust to peer influence, whereas some models show the opposite. Thus, there is no general guarantee that either system type produces more stable dynamics. Finally, our sensitivity analyses indicate that the content of the label-explanation pairs contributes to observed influence beyond the framing, with models differing in how they weight their own prior context versus external input. We hypothesise this is dependent on the quality of each model’s arguments.

In total, these findings suggest that influence in heterogeneous multi-agent systems follows a model-specific dynamic. Careful consideration of which models are included, what roles they play, and what tasks they are applied to is essential for building reliable and robust MAS.

Acknowledgements

We would like to thank Victoria Popa for her assistance with the sensitivity analyses and for running all experiments involving gpt-20b.

Computational Resources

Following climate-aware reporting practices, we report the total duration of all experiments. Across all experiments, including the baseline behaviour run, the agent interactions, and all sensitivity analyses, we have used approximately 170 GPU hours, excluding the gpt-20b runs. All computations were performed using the High Performance Cluster at the IT University of Copenhagen, located in Denmark.

Generative AI Statement

We have made use of generative AI for assistance for code debugging, refinement of plots and tables, and to provide support and inspiration for rephrasing.

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Appendix

A Counter Example: Index-based Pairing

In order to illustrate why an index based pairing is not optimal, consider two extreme scenarios. In the first scenario, both agents a^1 and a^2 produce identical label distributions across repetitions. They predict label 1 for repetitions $c = 0, \dots, 4$ and label 0 for $c = 5, \dots, 9$. Index-based matching would pair identical labels, resulting in perfect agreement. In the second scenario, agent a^1 retains the same sequence of outputs, while agent a^2 produces the same labels but in reversed order, predicting 0 for $c = 0, \dots, 4$ and 1 for $c = 5, \dots, 9$. Although the underlying predictive behaviour of a^2 is unchanged, index-based pairing now results in complete disagreement. This illustrates a need for a systematic pairing strategy that is invariant to repetition order and ensures that comparisons between agent pairs are based on the same foundation.

B Data Preprocessing

We apply preprocessing to the sarcasm and sentiment to remove low quality instances. For the CommonsenseQA 2.0 dataset, no additional filtering is applied, as the dataset is already well-curated benchmark of good quality. The preprocessing script is available at our Github repository at `src/preprocessing.py`.

B.1 Sarcasm

All claims of less than 6 words are omitted from the data, to ensure the instances keep a sufficient amount of context. Furthermore, we apply a keyword-based filter to remove claims containing strongly political topics, so the models are not affected by any potential guardrails. Further details on the filtering criteria are provided in the preprocessing script.

B.2 Sentiment

The sentiment dataset contains a number of semi-duplicate instances, due to its tree-based structure. In these cases, multiple claims may share most of the same text, while only a few words are added at a time. We therefore remove semi-duplicates when they have the same label, since these instances do not provide any additional information. However, we retain semi-duplicates when the added text changed the sentiment of the claim and therefore results in a different label.

For example, consider the following three instances:

Example of semi-duplicate instances		
25861,	The film boasts at least a few good ideas and features some decent performances,	<i>positive</i>
25862,	The film boasts at least a few good ideas and features some decent performances, but,	<i>positive</i>
25863,	The film boasts at least a few good ideas and features some decent performances, but the result is disappointing,	<i>negative</i>

In this example, the first two instances express the same sentiment and are therefore treated as semi-duplicates with identical labels. We remove the shorter instance and keep the longer one. The third instance is retained, since the added phrase changes the sentiment from positive to negative.

C Model Specifications

In Table 4, we provide further details about the models used in our experiments. Further details can be obtained by visiting a models respective Hugging Face repository.

MODEL NAME	PROVIDER	REPOSITORY ID	SAMPLING PARAMS.
gemma-3-4b	Google	unsloth/gemma-3-4b-it-unsloth-bnb-4bit	×
gemma-3-27b	Google	unsloth/gemma-3-27b-it-bnb-4bit	×
llama-3.1-8b	Meta	unsloth/Meta-Llama-3.1-8B-Instruct-bnb-4bit	✓
llama-3.3-70b	Meta	unsloth/Llama-3.3-70B-Instruct-bnb-4bit	✓
qwen-2.5-7b	Alibaba Cloud	unsloth/Qwen2.5-7B-Instruct-bnb-4bit	✓
qwen-2.5-72b	Alibaba Cloud	unsloth/Qwen2.5-72B-Instruct-bnb-4bit	✓
gpt-oss-20b	OpenAI	unsloth/gpt-oss-20b-unsloth-bnb-4bit	×

Table 4: Specification of models. Sampling params. refer to whether a generation configuration-file specifying default sampling parameters is provided in the repository. Further description of models without this is provided below.

C.1 Gemma: Sampling parameters

None of the two gemma models have any sampling parameters specified in their Hugging Face repositories, neither in the unsloth version or the repository for the original model.

Following sources, https://www.reddit.com/r/LocalLLaMA/comments/1j9hsfc/gemma_3_ggu_fs_recommended_settings/ and <https://docs.aimlapi.com/api-references/text-models-llm/google/gemma-3> discusses sampling parameters of the model, and we follow their recommendations.

C.2 Gpt: Sampling parameters

Gpt-20b does not specify sampling parameters in its Hugging Face repository. We therefore use the parameters from OpenAI’s official gpt-oss Github repository (<https://github.com/openai/gpt-oss/tree/main>), stated in the model’s README.

D Prompts

System prompt and user prompt structures for all datasets for both rounds.

System Prompt

Return ONLY a minified JSON object that conforms to this schema:

JSON schema for dataset

Rules:

- Output JSON only. No prose, no markdown, no extra text.
- Keys: label

Sarcasm User Prompts

R1

Is this claim sarcastic or literal?: **claim**
Provide an explanation with your reasoning.

R2

Is this claim sarcastic or literal?: **claim**
You previously said: **label**, with the following explanation: **explanation**.
However, a peer of yours labelled the claim as: **label**, with the following explanation: **explanation**. Having considered that, is this claim sarcastic or literal?: **claim**

: Receiver : Peer

Commonsense User Prompts

R1

Is this claim true?: **claim**
Provide an explanation with your reasoning.

R2

Is this claim true?: **claim**
You previously said: **label**, with the following explanation: **explanation**.
However, a peer of yours labelled the claim as: **label**, with the following explanation: **explanation**. Having considered that, is this claim true?: **claim**

: Receiver : Peer

Sentiment User Prompts

R1

Does this claim have positive or negative sentiment?: **claim**
Provide an explanation with your reasoning.

R2

Does this claim have positive or negative sentiment?: **claim**
You previously said: **label**, with the following explanation: **explanation**.
However, a peer of yours labelled the claim as: **label**, with the following explanation: **explanation**. Having considered that, does this claim have positive or negative sentiment?: **claim**

: Receiver : Peer

E Discarded Claims From Invalid Outputs

Table 5 shows the number of instances discarded from the first round outputs due to formatting that didn't conform with the expected JSON format. Given the small proportion of affected instances, they were discarded and therefore not included for interaction.

MODEL	SARCASM $N = 137,454$	COMMONSENSE $N = 9,264$	SENTIMENT $N = 91,964$
gemma-27b	4	1	11
gemma-4b	0	8	43
llama-70b	1	12	59
llama-8b	7	44	5
qwen-72b	27	1	0
qwen-7b	21	31	18
gpt-20b	0	0	0

Table 5: Number of instances discarded from baseline outputs per model and dataset. N denotes the total dataset size.

F Influence Heatmaps

We provide the additional influence heat maps for the two remaining datasets. The heat maps visualise the distribution of post-interaction certainty given the respective baseline certainty.

F.1 Commonsense

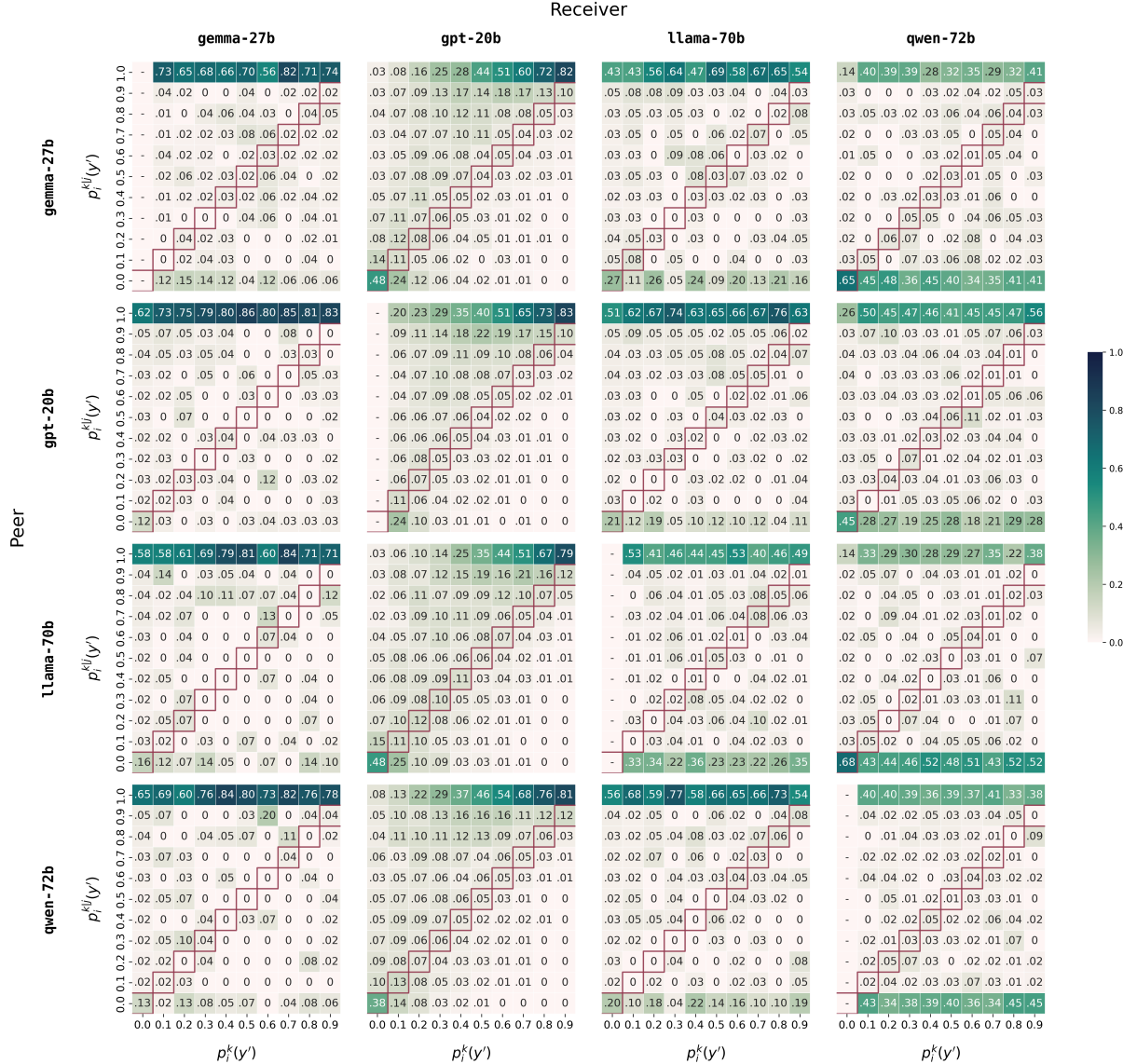


Figure 8: Conditional distribution of post-interaction certainty given baseline certainty for each receiver-peer pair on the **commonsense** dataset. A concentration of probability mass on the diagonal (marked with red) indicates that the receiver’s post-interaction certainty mirrors its baseline, suggesting stability under influence. Mass concentrated above the diagonal reflects positive influence, where mass below the diagonal are observations of negative influence. Plots on the diagonal correspond to homogeneous systems. The leftmost column in these plots is empty since only B:B match types are possible and thus excluding any instances where the baseline certainty is 0 or 1.

F.2 Sentiment

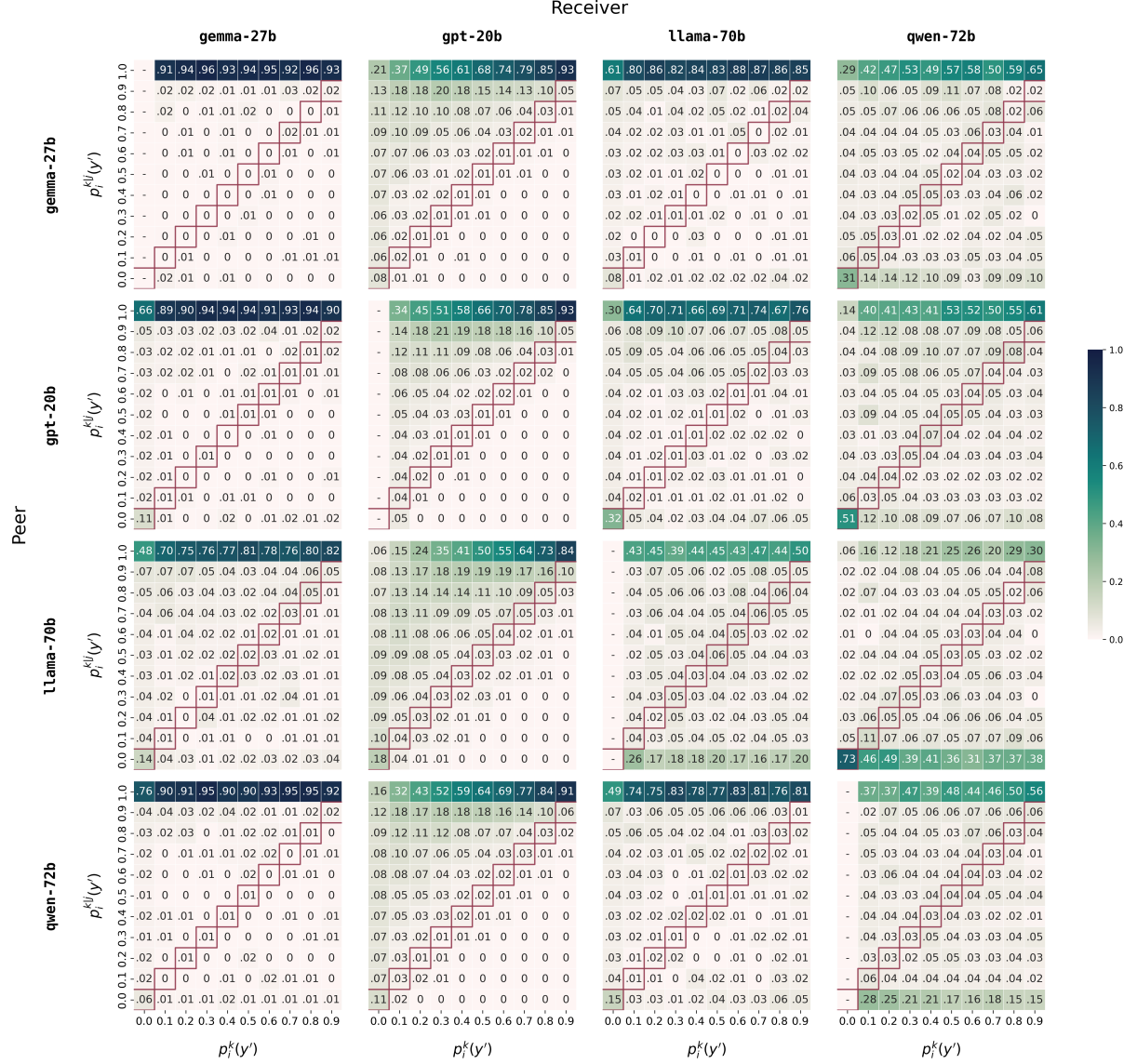


Figure 9: Conditional distribution of post-interaction certainty given baseline certainty for each receiver-peer pair on the **sentiment** dataset. A concentration of probability mass on the diagonal (marked with red) indicates that the receiver’s post-interaction certainty mirrors its baseline, suggesting stability under influence. Mass concentrated above the diagonal reflects positive influence, where mass below the diagonal are observations of negative influence. Plots on the diagonal correspond to homogeneous systems. The leftmost column in these plots is empty since only B:B match types are possible and thus excluding any instances where the baseline certainty is 0 or 1.

G Influence Scores on Agreeing Interactions

We report results of agreeing interaction configurations on the sarcasm dataset only, as these cases are less informative and observed behaviour is as expected. When the peer proposes a label that conforms with the opinion of the receiver, models remain stable.

All scores are based on instances within the following two match types: 0:0 and 1:1. Given these match types, the receiver assigns all probability mass to the peer’s proposed label already, leaving no room for positive influence. Thus, we can only observe an unexpected shift away from the agreement, i.e., any shift can only be observed in the negative influence score.

G.1 Large Models

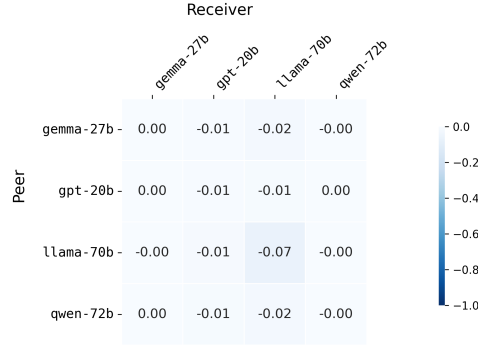


Figure 10: Negative influence scores across model pairs on the sarcasm dataset. All values are calculated from agreeing interactions, with consistent labels. Columns shows how susceptible a receiver is to influence from the different peers, while the rows show how much influence a peer can exert on different receivers.

G.2 Intra-Family

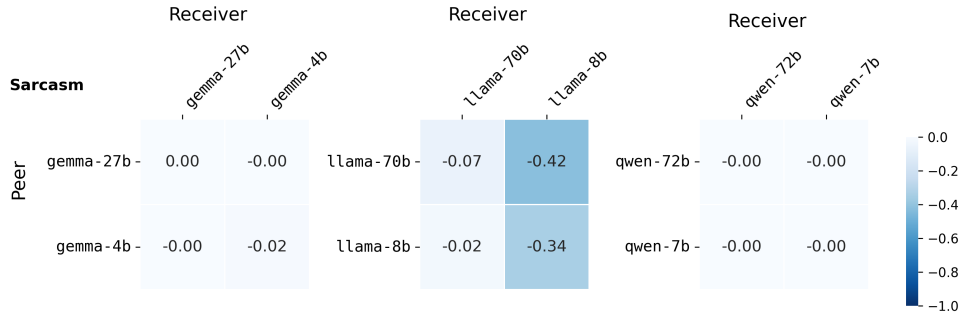


Figure 11: Intra-family negative influence scores on the sarcasm dataset. All values are calculated from agreeing interactions, with consistent labels. Columns shows how susceptible a receiver is to influence from the different peers, while the rows show how much influence a peer can exert on different receivers.

H Influence Differences Transitioning from Homogeneous to Heterogeneous

RECEIVER	SARCASM			COMMONSENSE			SENTIMENT		
	Hom. score	Het. avg.	%	Hom. score	Het. avg.	%	Hom. score	Het. avg	%
gemma-27b	0.97	0.86	-11.34	0.76	0.78	2.63	0.96	0.87	-9.38
gpt-20b	0.67	0.14	-79.1	0.57	0.42	-25.73	0.79	0.65	-17.30
llama-70b	0.51	0.58	13.73	0.54	0.69	27.16	0.56	0.71	26.19
qwen-72b	0.43	0.39	-9.30	0.45	0.42	-7.41	0.54	0.46	-15.43

Table 6: Comparison of susceptibility to influence in homogeneous (Hom.) and heterogeneous (Het.) system compositions, across all large receiver models on all datasets. For each receiver, the table reports the homogeneous positive influence score, the average positive influence score across heterogeneous interactions, and the percentage difference between the two. Positive differences indicates that the model becomes more susceptible to influence when transitioning from a homogeneous system to a heterogeneous system, while negative differences indicate that the model becomes less susceptible.

I Comparison of Heterogeneous and Homogeneous Systems

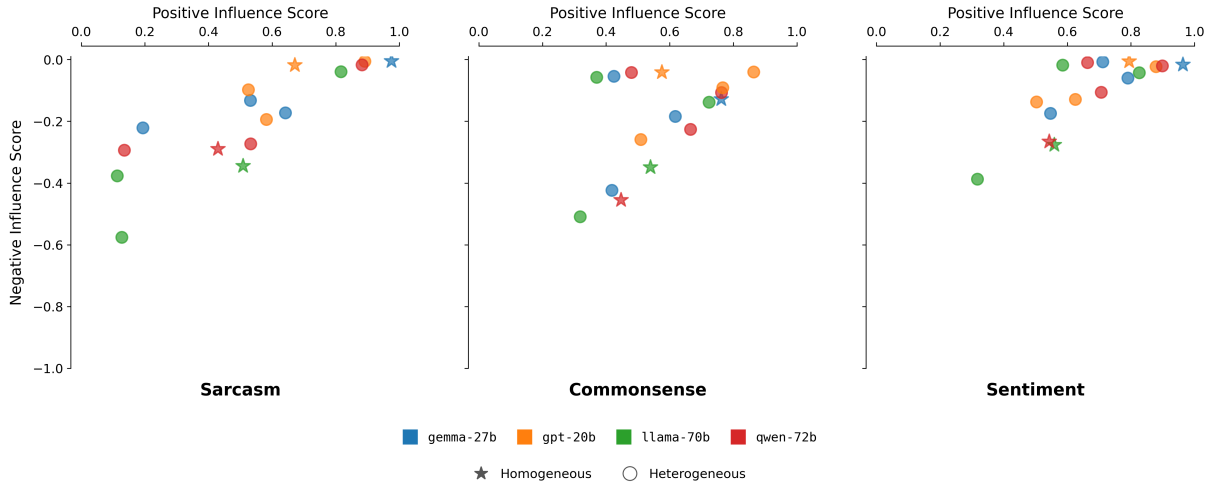


Figure 12: Influence patterns coloured by peer, with stars denoting homogeneous interactions and circles denoting heterogeneous interactions. Positive influence is shown along the x-axis, and the negative influence is shown along the y-axis. A non-influential peer will be placed close to the origin. If a peer lies in the upper-right region, it is able to highly influence a receiver towards its proposed label, whereas if it lies in the lower-left region, it induces strong backfiring tendencies in the receiver.

J Sensitivity Analysis - Prompt variation for Memoryless Receiver

R2

Is this claim sarcastic or literal?: **claim**
A peer of yours labelled the claim as: **label**,
with the following explanation: **explanation**.
Having considered that, is this claim sarcastic
or literal?: **claim**

: Peer

K Sensitivity Analysis - Prompt variation for No Explanations

R2

Is this claim sarcastic or literal?: **claim**
You previously said: **label**.
However, a peer of yours labelled the claim
as: **label**.
Having considered that, is this claim
sarcastic or literal?: **claim**

: Receiver : Peer

L Sensitivity Analysis - Prompt variation for Swap

R2

Is this claim sarcastic or literal?: **claim**
You previously said: **label**, with the
following explanation: **explanation**.
However, a peer of yours labelled the
claim as: **label**, with the following
explanation: **explanation**. Having
considered that, is this claim sarcastic
or literal?: **claim**

: Receiver : Peer

M Comparison of Model Explanations

We examine explanation quality more closely, motivated by the swap sensitivity analysis, where the opinions of the two agents are presented from reversed roles. The results of the analysis suggest that gpt-20b consistently produces more compelling explanations than llama-70b. Regardless of whether an explanation is framed as a peer opinion or the model’s own prior, both models show a systematic preference toward gpt-20b’s reasoning.

From a quantitative perspective, gpt-20b produces considerably longer explanations than llama-70b, with an average of 23 words per explanation compared to 7. This difference in length is also reflected in the quality of the explanations each model produces. With longer explanations gpt-20b is able to construct more elaborate arguments, whereas llama-70b tends to produce brief statements that assert a label without providing a supporting explanation (see examples below). This pattern is consistent across all interaction outcomes observed in the swap analysis. This suggests that explanation quality is a contributing factor to a peer’s ability to influence a receiver.

Model explanations	
llama-70b: <i>Speaker states a fact about learning something’s unconstitutionality.</i>	gpt-20b: <i>The sentence dismisses any benefit, highlighting absurdity. It critiques the claim’s premise, implying the speaker found the content unworthy or nonsense, typical sardonic tone.</i>
llama-70b: <i>Tone implies annoyance or frustration.</i>	gpt-20b: <i>The sentence is a straightforward inquiry about the availability of a release. It lacks the exaggerated tone, irony, or context typical of sarcasm.</i>
llama-70b: <i>Refers to past appearance as Marshmello</i>	gpt-20b: <i>The claim humorously implies someone claimed to be Marshmello, an unlikely identity, suggesting exaggeration or jest rather than fact.</i>

N Influence Scores on Match Type-Level

We report the positive and negative influence scores, and count of instances they are computed from, for each match type and dataset below.

RECEIVER ← PEER	MATCH TYPE	SARCASM			COMMONSENSE			SENTIMENT		
		IS ₊ ^{k j}	IS ₋ ^{k j}	Count	IS ₊ ^{k j}	IS ₋ ^{k j}	Count	IS ₊ ^{k j}	IS ₋ ^{k j}	Count
gemma-27b ← gemma-27b	0:0	–	0.00	1,000	–	-0.00	1,000	–	0.00	1,000
	1:1	–	0.00	1,000	–	-0.02	1,000	–	0.00	1,000
	B:B	0.97	-0.01	5,644	0.76	-0.13	482	0.96	-0.02	2,668
gemma-4b ← gemma-27b	0:0	–	-0.01	1,000	–	-0.00	1,000	–	–	–
	0:1	0.99	–	1,993	0.15	–	1,587	0.65	–	5,200
	0:B	0.97	–	305	0.09	–	156	0.60	–	846
	1:0	0.76	–	6,131	0.83	–	768	0.81	–	2,042
	1:1	–	-0.00	1,000	–	-0.03	1,000	–	–	–
	1:B	0.70	–	2,439	0.87	–	67	0.75	–	419
	B:0	0.95	-0.02	411	0.93	-0.06	187	0.90	-0.05	396
	B:1	0.99	-0.00	713	0.22	-0.66	288	0.86	-0.07	745
	B:B	0.97	-0.00	156	0.38	-0.32	34	0.85	-0.10	136
gpt-20b ← gemma-27b	0:0	–	-0.00	1,000	–	–	–	–	–	–
	0:1	0.19	–	7,000	0.19	–	600	0.53	–	952
	0:B	0.10	–	1,857	0.14	–	99	0.57	–	277
	1:0	0.18	–	18	0.37	–	75	0.71	–	972
	1:1	–	-0.01	1,000	–	–	–	–	–	–
	1:B	0.12	–	15	0.56	–	20	0.65	–	254
	B:0	0.21	-0.26	1,811	0.71	-0.03	1,026	0.91	-0.00	5,989
	B:1	0.34	-0.14	7,000	0.50	-0.06	2,162	0.81	-0.01	7,000
	B:B	0.21	-0.27	1,900	0.51	-0.07	244	0.80	-0.01	1,606
llama-70b ← gemma-27b	0:0	–	-0.01	1,000	–	-0.02	1,000	–	–	–
	0:1	0.87	–	7,000	0.51	–	842	0.66	–	710
	0:B	0.79	–	2,036	0.37	–	126	0.64	–	224
	1:0	0.33	–	421	0.72	–	480	0.84	–	4,394
	1:1	–	-0.03	1,000	–	-0.04	1,000	–	–	–

continued...

RECEIVER $\leftarrow$ PEER	MATCH TYPE	SARCASM			COMMONSENSE			SENTIMENT		
		$IS_{+}^{k j}$	$IS_{-}^{k j}$	Count	$IS_{+}^{k j}$	$IS_{-}^{k j}$	Count	$IS_{+}^{k j}$	$IS_{-}^{k j}$	Count
	1:B	0.35	–	398	0.77	–	100	0.77	–	1,033
	B:0	0.47	-0.32	524	0.85	-0.09	114	0.95	-0.02	970
	B:1	0.96	-0.01	7,000	0.57	-0.33	188	0.77	-0.12	273
	B:B	0.72	-0.19	776	0.54	-0.14	30	0.91	-0.04	154
qwen-72b $\leftarrow$ gemma-27b	0:0	–	-0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.40	–	7,000	0.11	–	1,001	0.41	–	2,097
	0:B	0.26	–	1,547	0.12	–	146	0.34	–	592
	1:0	0.60	–	1,123	0.59	–	379	0.64	–	1,723
	1:1	–	-0.00	1,000	–	-0.03	1,000	–	–	–
	1:B	0.56	–	881	0.63	–	72	0.49	–	589
	B:0	0.76	-0.06	749	0.78	-0.12	134	0.74	-0.12	654
	B:1	0.57	-0.16	5,797	0.19	-0.63	285	0.61	-0.20	788
	B:B	0.58	-0.18	788	0.51	-0.52	46	0.60	-0.21	306
gemma-27b $\leftarrow$ gemma-4b	0:0	–	0.00	1,000	–	-0.01	1,000	–	–	–
	0:1	0.90	–	6,131	0.38	–	768	0.58	–	2,042
	0:B	0.84	–	411	0.31	–	187	0.58	–	396
	1:0	0.59	–	1,993	0.51	–	1,587	0.89	–	5,200
	1:1	–	-0.00	1,000	–	-0.02	1,000	–	–	–
	1:B	0.57	–	713	0.48	–	288	0.81	–	745
	B:0	0.88	-0.05	305	0.71	-0.15	156	0.98	-0.00	846
	B:1	0.95	-0.01	2,439	0.60	-0.37	67	0.88	-0.07	419
	B:B	0.92	-0.01	156	0.56	-0.24	34	0.97	-0.03	136
gemma-4b $\leftarrow$ gemma-4b	0:0	–	-0.03	1,000	–	-0.00	1,000	–	0.00	1,000
	1:1	–	-0.00	1,000	–	-0.03	1,000	–	-0.00	1,000
	B:B	0.90	-0.04	2,404	0.40	-0.48	984	0.63	-0.22	2,418
gemma-27b $\leftarrow$ gpt-20b	0:0	–	0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.97	–	18	0.94	–	75	0.75	–	972
	0:B	0.98	–	1,811	0.76	–	1,026	0.67	–	5,989
	1:0	0.83	–	7,000	0.92	–	600	0.96	–	952
	1:1	–	0.00	1,000	–	-0.00	1,000	–	–	–
	1:B	0.54	–	7,000	0.73	–	2,162	0.89	–	7,000
	B:0	0.97	-0.00	1,857	0.94	-0.02	99	1.00	0.00	277
	B:1	0.99	0.00	15	0.94	0.00	20	0.93	-0.05	254
	B:B	0.96	-0.01	1,900	0.83	-0.10	244	0.95	-0.02	1,606
gpt-20b $\leftarrow$ gpt-20b	0:0	–	-0.01	1,000	–	–	–	–	–	–
	1:1	–	-0.01	1,000	–	–	–	–	–	–
	B:B	0.67	-0.02	14,000	0.57	-0.04	6,620	0.79	-0.01	14,000
llama-70b $\leftarrow$ gpt-20b	0:0	–	-0.02	1,000	–	-0.00	1,000	–	–	–
	0:1	0.90	–	267	0.85	–	97	0.45	–	548
	0:B	0.86	–	7,000	0.59	–	1,210	0.40	–	3,736
	1:0	0.25	–	7,000	0.86	–	415	0.72	–	2,393
	1:1	–	-0.00	1,000	–	-0.00	1,000	–	–	–
	1:B	0.11	–	7,000	0.67	–	1,914	0.48	–	7,000
	B:0	0.37	-0.35	3,742	0.89	-0.04	111	0.90	-0.04	596
	B:1	0.97	-0.00	345	0.79	-0.06	22	0.61	-0.26	139
	B:B	0.61	-0.22	12,432	0.72	-0.18	368	0.81	-0.09	1,170
qwen-72b $\leftarrow$ gpt-20b	0:0	–	0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.63	–	81	0.38	–	76	0.35	–	1,006
	0:B	0.43	–	5,128	0.16	–	1,327	0.24	–	6,151
	1:0	0.39	–	7,000	0.81	–	245	0.60	–	752
	1:1	–	0.00	1,000	–	-0.00	1,000	–	–	–
	1:B	0.14	–	7,000	0.55	–	1,710	0.33	–	7,000
	B:0	0.70	-0.07	3,305	0.91	-0.03	123	0.77	-0.06	361
	B:1	0.79	-0.07	110	0.26	-0.37	47	0.62	-0.18	291
	B:B	0.59	-0.15	7,050	0.49	-0.37	544	0.61	-0.17	1,886
gemma-27b $\leftarrow$ llama-70b	0:0	–	0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.94	–	421	0.72	–	480	0.67	–	4,394
	0:B	0.92	–	524	0.58	–	114	0.56	–	970
	1:0	0.64	–	7,000	0.75	–	842	0.94	–	710
	1:1	–	-0.00	1,000	–	-0.01	1,000	–	–	–
	1:B	0.51	–	7,000	0.71	–	188	0.93	–	273
	B:0	0.85	-0.06	2,036	0.85	-0.10	126	0.99	-0.00	224
	B:1	0.95	-0.01	398	0.72	-0.15	100	0.83	-0.08	1,033
	B:B	0.89	-0.05	776	0.73	-0.17	30	0.88	-0.04	154
gpt-20b $\leftarrow$ llama-70b	0:0	–	-0.01	1,000	–	–	–	–	–	–
	0:1	0.14	–	7,000	0.18	–	415	0.39	–	2,393
	0:B	0.12	–	3,742	0.11	–	111	0.38	–	596
	1:0	0.04	–	267	0.37	–	97	0.60	–	548
	1:1	–	-0.02	1,000	–	–	–	–	–	–
	1:B	0.04	–	345	0.42	–	22	0.56	–	139
	B:0	0.06	-0.48	7,000	0.61	-0.04	1,210	0.85	-0.01	3,736
	B:1	0.25	-0.21	7,000	0.46	-0.06	1,914	0.65	-0.02	7,000
	B:B	0.14	-0.44	12,432	0.45	-0.08	368	0.68	-0.02	1,170
llama-70b $\leftarrow$ llama-70b	0:0	–	-0.08	1,000	–	-0.02	1,000	–	-0.00	1,000

continued...

RECEIVER ← PEER	MATCH TYPE	SARCASM			COMMONSENSE			SENTIMENT		
		$IS_+^{k j}$	$IS_-^{k j}$	Count	$IS_+^{k j}$	$IS_-^{k j}$	Count	$IS_+^{k j}$	$IS_-^{k j}$	Count
	1:1	–	-0.05	1,000	–	-0.05	1,000	–	-0.01	1,000
	B:B	0.51	-0.34	14,000	0.54	-0.35	634	0.56	-0.28	2,640
llama-8b ← llama-70b	0:0	–	-0.83	369	–	-0.42	1,000	–	-0.00	1,000
	0:1	0.84	–	20	0.59	–	77	0.31	–	1,095
	0:B	0.86	–	7	0.47	–	13	0.31	–	331
	1:0	0.00	–	7,000	0.66	–	1,071	0.26	–	1,697
	1:1	–	-0.02	1,000	–	-0.20	1,000	–	-0.00	1,000
	1:B	0.00	–	7,000	0.60	–	152	0.28	–	415
	B:0	0.00	-0.91	7,000	0.63	-0.07	2,731	0.20	-0.36	4,044
	B:1	0.93	-0.00	6,326	0.36	-0.25	1,126	0.25	-0.31	3,630
	B:B	0.25	-0.27	4,748	0.48	-0.22	292	0.20	-0.36	1,148
qwen-72b ← llama-70b	0:0	–	-0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.10	–	2,404	0.08	–	715	0.11	–	4,578
	0:B	0.07	–	1,738	0.04	–	174	0.05	–	968
	1:0	0.13	–	7,000	0.46	–	465	0.45	–	588
	1:1	–	0.00	1,000	–	-0.02	1,000	–	–	–
	1:B	0.08	–	7,000	0.50	–	113	0.40	–	249
	B:0	0.23	-0.47	3,266	0.67	-0.22	167	0.67	-0.17	216
	B:1	0.14	-0.63	2,471	0.14	-0.79	242	0.23	-0.56	1,275
	B:B	0.14	-0.63	2,406	0.34	-0.52	60	0.31	-0.43	206
llama-70b ← llama-8b	0:0	–	-0.02	369	–	-0.01	1,000	–	-0.00	1,000
	0:1	0.67	–	7,000	0.23	–	1,071	0.33	–	1,697
	0:B	0.59	–	7,000	0.15	–	2,731	0.20	–	4,044
	1:0	0.15	–	20	0.47	–	77	0.58	–	1,095
	1:1	–	-0.03	1,000	–	-0.05	1,000	–	-0.01	1,000
	1:B	0.09	–	6,326	0.53	–	1,126	0.48	–	3,630
	B:0	0.33	-0.44	7	0.58	-0.36	13	0.77	-0.10	331
	B:1	0.89	-0.03	7,000	0.33	-0.43	152	0.46	-0.32	415
	B:B	0.57	-0.32	4,748	0.48	-0.33	292	0.57	-0.24	1,148
llama-8b ← llama-8b	0:0	–	-0.68	396	–	-0.30	1,000	–	-0.00	1,000
	1:1	–	-0.01	1,000	–	-0.17	1,000	–	-0.00	1,000
	B:B	0.26	-0.28	14,000	0.50	-0.20	8,014	0.29	-0.28	14,000
gemma-27b ← qwen-72b	0:0	–	0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.98	–	1,123	0.77	–	379	0.78	–	1,723
	0:B	0.95	–	749	0.61	–	134	0.75	–	654
	1:0	0.76	–	7,000	0.82	–	1,001	0.95	–	2,097
	1:1	–	0.00	1,000	–	-0.01	1,000	–	–	–
	1:B	0.65	–	5,797	0.70	–	285	0.94	–	788
	B:0	0.92	-0.03	1,547	0.81	-0.07	146	0.99	-0.00	592
	B:1	0.98	-0.00	881	0.78	-0.12	72	0.92	-0.05	589
	B:B	0.93	-0.02	788	0.83	-0.13	46	0.97	-0.01	306
gpt-20b ← qwen-72b	0:0	–	-0.00	1,000	–	–	–	–	–	–
	0:1	0.14	–	7,000	0.27	–	245	0.53	–	752
	0:B	0.10	–	3,305	0.20	–	123	0.55	–	361
	1:0	0.07	–	81	0.53	–	76	0.59	–	1,006
	1:1	–	-0.02	1,000	–	–	–	–	–	–
	1:B	0.07	–	110	0.49	–	47	0.53	–	291
	B:0	0.13	-0.34	5,128	0.74	-0.02	1,327	0.85	-0.01	6,151
	B:1	0.27	-0.18	7,000	0.58	-0.04	1,710	0.81	-0.01	7,000
	B:B	0.16	-0.35	7,050	0.54	-0.07	544	0.79	-0.02	1,886
llama-70b ← qwen-72b	0:0	–	-0.01	1,000	–	-0.01	1,000	–	–	–
	0:1	0.80	–	7,000	0.63	–	465	0.61	–	588
	0:B	0.73	–	3,266	0.47	–	167	0.53	–	216
	1:0	0.20	–	2,404	0.78	–	715	0.73	–	4,578
	1:1	–	-0.03	1,000	–	-0.02	1,000	–	–	–
	1:B	0.17	–	2,471	0.65	–	242	0.62	–	1,275
	B:0	0.31	-0.50	1,738	0.85	-0.09	174	0.91	-0.04	968
	B:1	0.92	-0.03	7,000	0.58	-0.30	113	0.73	-0.18	249
	B:B	0.59	-0.30	2,406	0.70	-0.29	60	0.82	-0.09	206
qwen-72b ← qwen-72b	0:0	–	-0.00	1,000	–	-0.00	1,000	–	0.00	1,000
	1:1	–	-0.00	1,000	–	-0.02	1,000	–	0.00	1,000
	B:B	0.43	-0.29	13,880	0.45	-0.45	884	0.54	-0.27	3,190
qwen-7b ← qwen-72b	0:0	–	-0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.75	–	128	0.26	–	1,830	0.56	–	1,371
	0:B	0.77	–	95	0.18	–	372	0.52	–	528
	1:0	0.19	–	7,000	0.89	–	168	0.55	–	3,472
	1:1	–	-0.00	1,000	–	-0.02	1,000	–	–	–
	1:B	0.11	–	6,790	0.80	–	53	0.48	–	911
	B:0	0.28	-0.42	465	0.84	-0.13	75	0.62	-0.17	993
	B:1	0.78	-0.12	85	0.31	-0.50	203	0.61	-0.22	575
	B:B	0.52	-0.27	110	0.57	-0.24	32	0.49	-0.24	310
qwen-72b ← qwen-7b	0:0	–	-0.00	1,000	–	-0.00	1,000	–	–	–
	0:1	0.11	–	7,000	0.06	–	168	0.16	–	3,472
	0:B	0.05	–	465	0.05	–	75	0.13	–	993

continued...

RECEIVER $\leftarrow$ PEER	MATCH TYPE	SARCASM			COMMONSENSE			SENTIMENT		
		$IS_+^{k j}$	$IS_-^{k j}$	Count	$IS_+^{k j}$	$IS_-^{k j}$	Count	$IS_+^{k j}$	$IS_-^{k j}$	Count
	1:0	0.18	–	128	0.32	–	1,830	0.35	–	1,371
	1:1	–	-0.00	1,000	–	-0.02	1,000	–	–	–
	1:B	0.15	–	85	0.26	–	203	0.24	–	575
	B:0	0.27	-0.37	95	0.51	-0.29	372	0.50	-0.28	528
	B:1	0.18	-0.50	6,790	0.11	-0.85	53	0.35	-0.42	911
	B:B	0.21	-0.55	110	0.34	-0.56	32	0.34	-0.44	310
qwen-7b $\leftarrow$ qwen-7b	0:0	–	-0.00	1,000	–	-0.02	1,000	–	-0.00	1,000
	1:1	–	-0.00	1,000	–	-0.03	1,000	–	-0.00	1,000
	B:B	0.42	-0.33	1,210	0.40	-0.40	588	0.47	-0.29	3,446

Table 7: Positive and negative influence scores, reported together with the corresponding number of instances from which they were computed. ‘–’ indicates match types for which the score cannot be computed, as either no positive or negative shift can occur given the baseline label certainty of the receiver model. Recall that 0:0 and 1:1 are agreeing configurations, and the remaining match types describe disagreeing configurations.